Satellite Link Budget Tool

User Manual

Software version 1.4.0

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1 Introduction

The Satellite Link Budget Tool computes the radio link budget between a low Earth orbit satellite and a ground station, for the downlink (satellite to ground) and the uplink (ground to satellite). It offers two complementary analyses:

- **Pass analysis**: from the satellite TLE, the tool propagates the orbit over a whole day, finds the passes (contact windows) over the selected ground station and computes, every 30 seconds of each pass, the elevation, slant range, Doppler shift and the downlink and uplink link budgets.
- **Static link budget**: the classic preliminary, worst-case check at a single fixed elevation (typically the minimum operational one), independent of any TLE, laid out as an ECSS static link budget sheet with clear-sky and rain-faded conditions, statistical margins from favourable/adverse tolerances, the power flux density (PFD) regulatory check and a PDF report.

Both analyses use the same models: ITU-R atmospheric attenuation, ionospheric scintillation from an S4 index, polarisation mismatch, the satellite antenna pattern and the CCSDS telemetry framing, so that their results can be compared.

1.1 Conventions

- Names of buttons, fields and tabs are written in **bold**, as they appear on screen.
- Values are in dB, dBW, dBi, dB/K, dBHz unless stated; angles in degrees; bit rates as labelled.
- Every input field has an **(i)** icon next to it: hovering over it shows what the field means and how it is used. Every row of the result tables shows its definition when hovered.

2 Installation and start

2.1 Program folder

The program is delivered as a folder, **Link-Budget-Tool**, usually as a zip file. Unzip it anywhere (e.g. C:\Programs\Exostaff\Link-Budget-Tool); no Python installation is needed. The folder contains **Link-Budget-Tool.exe**, its libraries (in _internal) and the data files below, which the program reads at startup and which can be edited or replaced without rebuilding it:

File	Content
`license.key`	Licence of the installation (see 2.2). Required when the build checks licences.
`ground_stations.csv`	Ground station catalogue (CSV with header, opens in Excel): name, latitude, longitude, altitude (m); optional antenna diameter (m), own G/T, its elevation, Tsys and number of antennas. Edit Stations... edits it as a table.
`satellites.json`	The satellites of the Satellite menu (chapter 4.1).
`missions\\`	Per mission: parameters.json (project name, link budget parameters, static inputs; chapter 12), antenna_pattern.txt and tle.txt (chapter 4.1).

A missing data file is not an error: the related fields stay empty (or the built-in station list and antenna pattern are used) and can be filled in, or loaded from another file, in the program.

2.2 Licence

The executable can be time-limited. The licence is the file **license.key**, placed next to Link-Budget-Tool.exe; it names the customer and the expiry date and is signed, so it cannot be edited. A valid licence is shown at the top right of the window ('Licensed to … · valid until …', in orange during its last 30 days). Without a valid licence the program stops at startup with one of these messages:

Message	What to do
No licence found	Copy the license.key you received into the program folder.
The licence expired on …	Ask the supplier for a new licence file.
The licence file is not valid / damaged	The file was edited or corrupted: use the original file.
The system date appears to have been set back	Correct the computer date and restart.

2.3 Starting the program

Double-click **Link-Budget-Tool.exe**. A window with the Exostaff logo appears immediately and shows the progress while the program loads (a few seconds). The main window then opens with the data files already loaded. Close the program with **Exit** (top right), Ctrl+Q or the window close button.

3 Main window

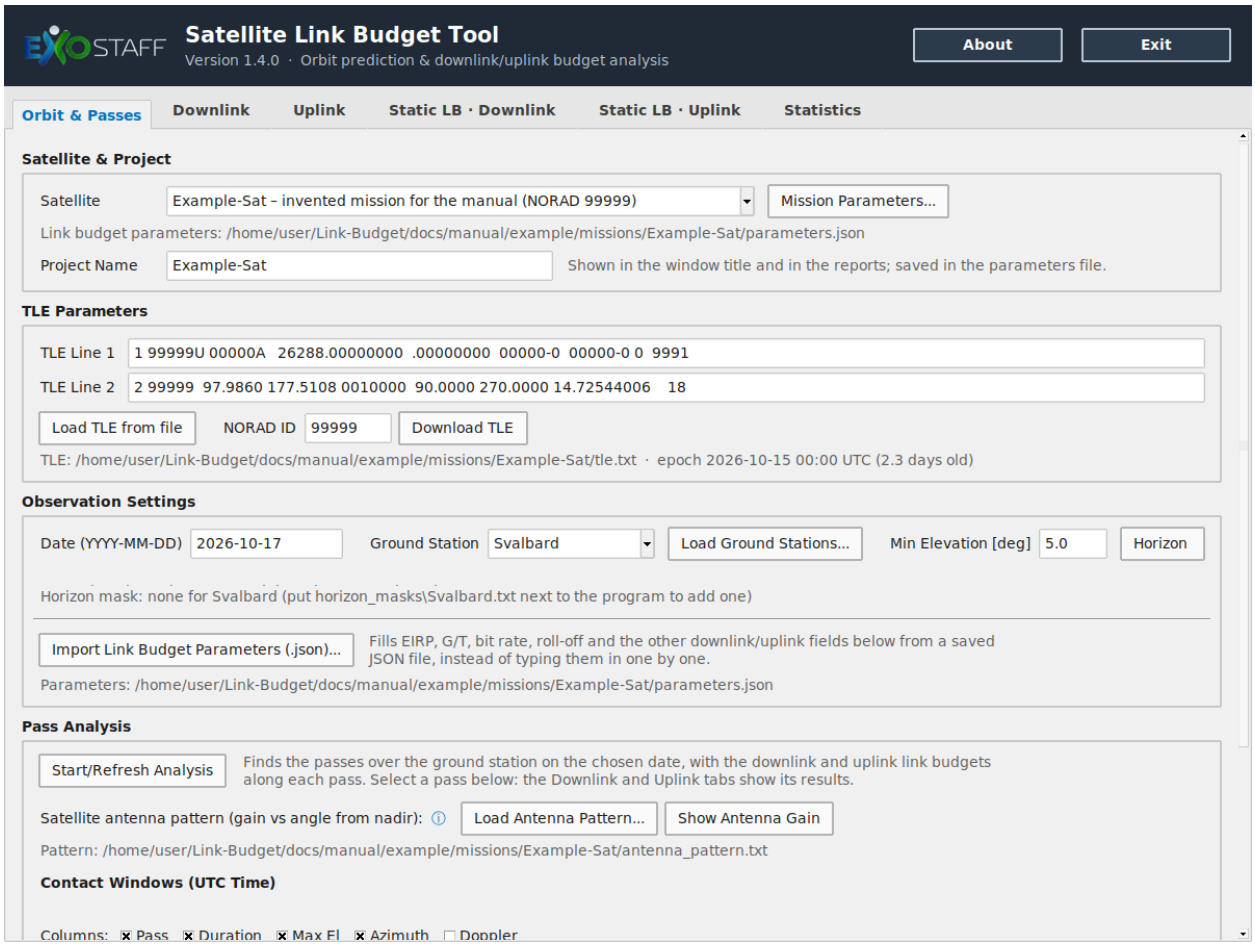


Figure 1 – Main window, Orbit & Passes tab.

The header shows the program version, the licence and the **About** (version, author and licence; also F1) and **Exit** buttons. Below it, the tabs:

Tab	Content
Orbit & Passes	Project name, TLE, date and ground station, data files, the pass analysis and the list of passes (chapter 4).
Statistics	Contacts, data volume, on-board memory and latency over several days, stations and satellites (chapter 5).
Downlink	Downlink parameters, and plots and table of the selected pass (chapter 6).
Uplink	Uplink parameters and table of the selected pass (chapter 7).
Static LB · Downlink	Fixed-elevation downlink link budget (chapter 8).
Static LB · Uplink	Fixed-elevation uplink link budget (chapter 8).

Shared fields. Frequency, EIRP, G/T, bit rate, overhead, roll-off, spectral efficiency, demodulator loss, other attenuations, antenna diameter and link availability belong to the Downlink and Uplink tabs. The static tabs show them again with a **blue label**: they are the same fields, so editing one changes the other.

A button that needs to be pressed again because its inputs changed is shown in **blue** (Start/Refresh Analysis, Calculate UL Link Budget, Calculate in the static tabs).

4 Orbit & Passes tab

4.1 Satellite and project

Satellite lists the satellites of **satellites.json** (EUMETSAT missions; the file can be edited). Each satellite has a folder **missions/<name>** next to the program:

- **parameters.json**: the link budget parameters of the mission (all the fields of the Downlink, Uplink and Static LB tabs). When it is present they are loaded and the link budget is available. **Mission Parameters…**, next to the Satellite menu, shows the whole file as a table grouped by subject (mission, orbit, downlink, ground station, uplink, statistics, static link budgets), with units and descriptions: double-click a value to change it (an empty value removes it), Save writes the file and loads the values in the window.
- Without parameters.json only the **contact analysis** is done: the passes with their times, duration and maximum elevation. The Downlink, Uplink and Static LB tabs are disabled. To add the link budget, copy the parameters.json of a similar mission into the folder and edit it.
- **antenna_pattern.txt** (optional; or any .txt/.csv file with "pattern" in its name): the satellite antenna pattern of the mission (two columns, angle from nadir in degrees and gain in dBi, separated by commas, tabs, spaces or semicolons, then also with a decimal comma as Excel writes it with Italian settings; header lines and # comments are skipped). There is no common default: a mission without a readable pattern gets an isotropic 0 dBi antenna, reported in orange. The line under the pattern buttons and the title of Show Antenna Gain give the file in use; Load Antenna Pattern… only replaces it for the session.
- **tle.txt**: the last TLE downloaded (or, for a custom orbit, the last synthetic TLE generated). On selection the TLE is downloaded again when it is older than one day; without internet the saved one is used.

A mission not launched yet has a **custom orbit**, given by its mean elements: **altitude** (mean semi-major axis minus the equatorial radius), **eccentricity**, argument of perigee **ω** (90° for a frozen orbit), **inclination** (empty: computed for a sun-synchronous orbit), on the **Custom orbit** row. The **Constellation** row gives the orbital **planes** by the local time of their ascending node (LTAN, hh:mm separated by commas, e.g. 15:30, 19:30, 23:30), the satellites per plane (**Sats/plane**, evenly spaced in the plane) and the **Phasing** (shift of the argument of latitude from one plane to the next); **Extra sats** adds single satellites to the regular ones, written LTAN@argument of latitude and separated by commas (e.g. 19:30@120; a new LTAN makes a plane of its own). **Analyse: plane / sat #** chooses the satellite used by the pass analysis and the link budgets. The program builds a synthetic TLE with its epoch at the analysed date (also with **Generate TLE**), with the mean motion set so that the nodal period is that of the J2 theory: the pass times do not drift over the day. Passes and link budgets are then computed as for a launched satellite. The example constellation of the figures (Example-Constellation, an invented mission) has 650 km, e = 0.001, ω = 90°, inclination computed for a sun-synchronous orbit (97.986°), planes at 10:30 and 22:30, 3 satellites per plane, phasing 60°. SGP4 adds its short-period terms to the mean elements, so the instantaneous altitude swings a few km around the mean one; the mean altitude and the pass times are those of the mean elements. The orbit is saved in the orbit section of the mission's parameters.json with **Save** (Static LB tabs); the Statistics tab can use the whole constellation. To change the orbits: edit the fields (a new plane list or Sats/plane updates the plane and satellite menus), check them with **Show Constellation**, then Save.

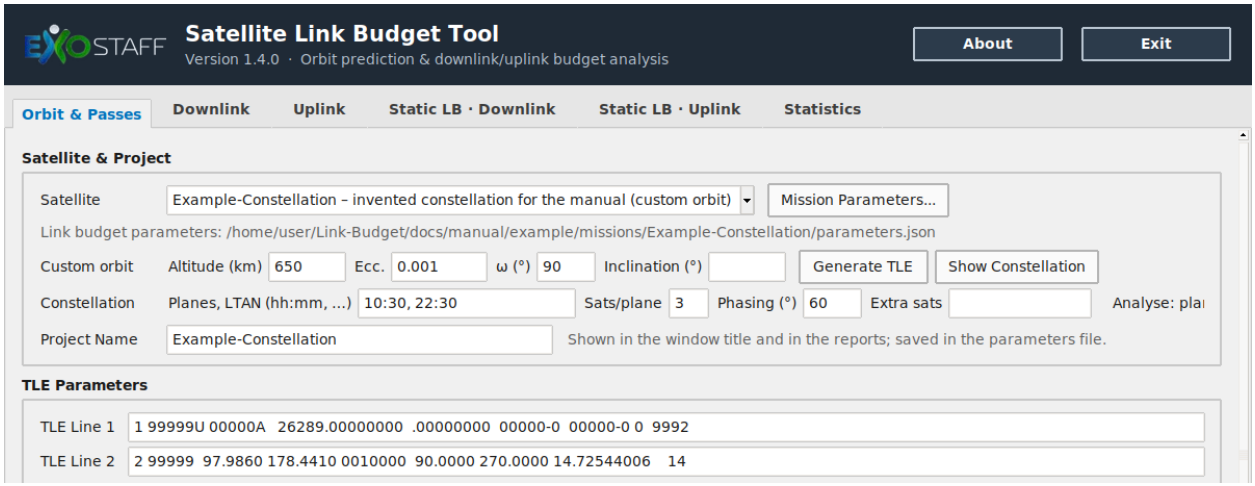


Figure 2 – Example-Constellation selected: custom orbit, constellation and synthetic TLE.

Show Constellation lists every satellite (plane, RAAN, argument of latitude and inclination at 00:00 UTC of the analysed date) and draws the ground tracks over the first 100 minutes of the day with the ground stations, and the satellites in their planes (RAAN against argument of latitude). **3D View** draws the orbits around the Earth (inertial frame), the satellites and the stations; drag to rotate, and move the slider to follow the satellites over the first 100 minutes of the day.

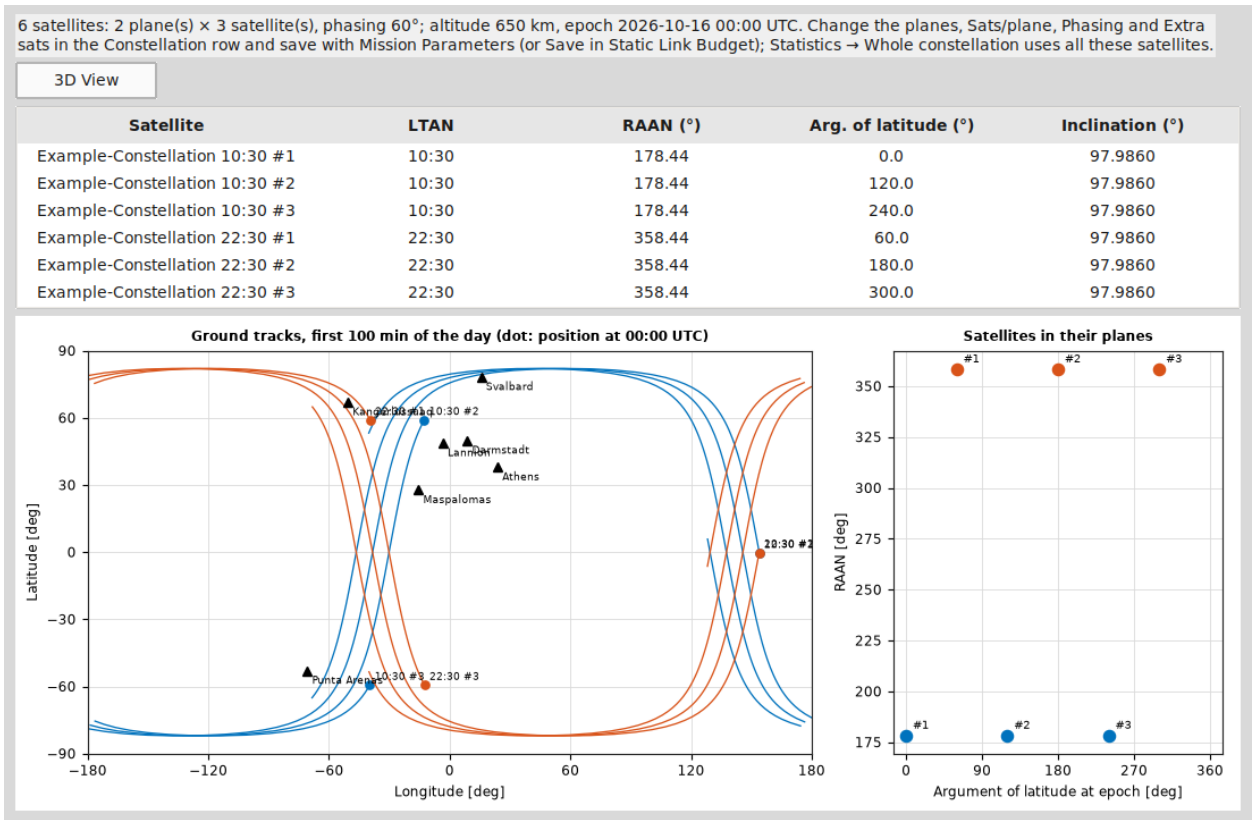


Figure 3 – Show Constellation: 2 planes × 3 satellites, phasing 60°.

Project Name is shown in the window title, in the PDF report (page header and above the title), in the proposed PDF file name and in the CSV exports. It is read from and saved to the parameters file (key project_name).

4.2 TLE and observation settings

- **TLE Line 1 / Line 2**: the satellite elements, filled from tle.txt at startup or with **Load TLE from file**. The line below shows the file, the TLE epoch and its age; an epoch older than 7 days is shown in orange, because the pass times drift.
- **NORAD ID** and **Download TLE**: download the latest TLE of the satellite and save it in tle.txt. The sources are tried in turn until one answers: SatNOGS (db.satnogs.org, elements from Space-Track; it only lists satellites with a known radio transmitter) and N2YO (the TLE on the satellite's page). When both fail, the status line gives the reason for each source. The NORAD ID is read from and saved to the parameters file (key norad_id). Without internet access the TLE already loaded is kept: copy the TLE from a web page into tle.txt instead.
- **Date** (YYYY-MM-DD): the day analysed (00:00–24:00 UTC, 30 s step).
- **Ground Station**: the station of the pass analysis, from the catalogue (ground_stations.csv); **Load Ground Stations…** reads another catalogue and **Edit Stations…** shows the catalogue as a table to add, change or delete stations (position, diameter, own G/T and Tsys, antennas) and save it. The line below shows which file is in use.
- **Min Elevation [deg]** (default 5°) and the station's **horizon mask**: a sample is in contact only above both. The mask is the file horizon_masks\<station>.txt next to the program (two columns: azimuth from North and minimum elevation, linear between the points; see horizon_masks\README.txt); **Horizon Mask** plots it, and the line under the station gives the file in use. Without a file only the minimum elevation applies.
- **Import Link Budget Parameters (.json)…**: fills all the parameters, the project name and the static tabs from a parameters file.

4.3 Pass analysis

Start/Refresh Analysis propagates the orbit for the chosen date and station, finds the passes above the minimum elevation (5°) and computes the downlink and uplink link budgets along each of them. The passes are listed in the **Contact Windows (UTC Time)** table, one row per pass: start and end (UTC), direction (Asc. or Desc.), duration, maximum elevation, azimuth at the start, at the maximum elevation and at the end (degrees from North clockwise, with the compass direction: where to point the antenna) and, optionally, the largest Doppler shift and rate. The **Columns** check boxes choose the columns shown; hover a heading for its meaning. A pass that exceeds the PFD limit is red, one with noise from the Sun orange; hovering the row gives the detail (by how much the limit is exceeded, the Sun noise and its time). **Simultaneous with station** chooses a second station, e.g. the one the satellite dumps its data to: the table then gives, for each contact, the minutes during which the satellite sees both stations and that common window. The field **above … ° there** gives the elevation above which the satellite counts as seen by the second station, e.g. the elevation above which it transmits to it (3°); empty, the Min Elevation is used; the horizon mask of the second station is used too. A satellite that transmits only over its own station can be received here only in those minutes. With **Analyse the common minutes only** the contacts become those common windows (passes with none disappear): plots, tables and link budgets of the Downlink and Uplink tabs cover only them. Select a pass: the Downlink and Uplink tabs show its results. The button turns blue when the TLE, date or station change.

Satellite antenna pattern: the gain of the satellite antenna versus the angle from nadir, used at every time step (chapter 9.4). It is read from the folder of the selected mission; **Load Antenna Pattern…** loads another file for the session and **Show Antenna Gain** plots it. The line below the buttons shows which pattern is in use.

5 Statistics tab

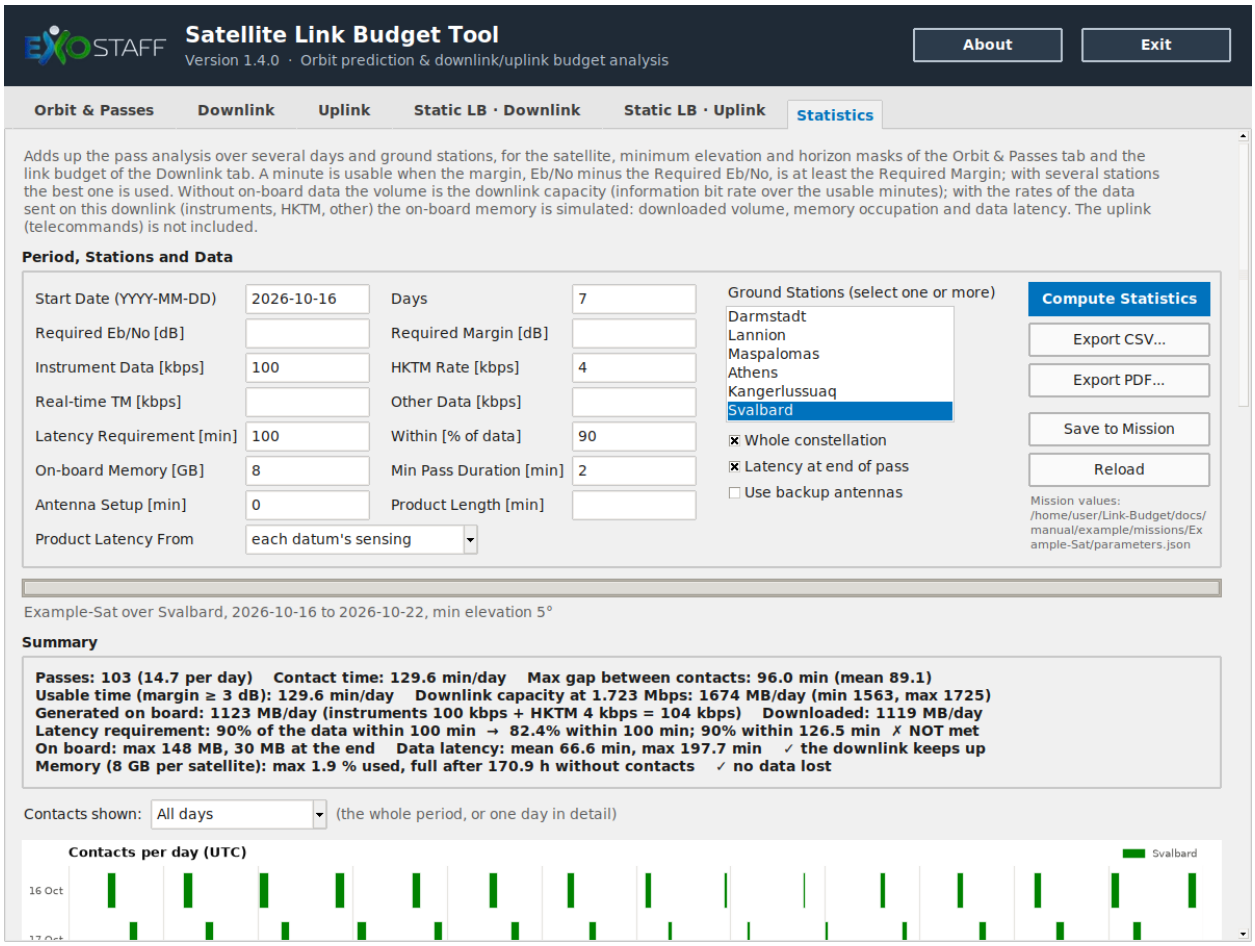


Figure 4 – Statistics over 7 days: summary and data per day.

The **Statistics** tab adds up the pass analysis over several days and ground stations, for the satellite, minimum elevation and horizon masks of the Orbit & Passes tab and the link budget of the Downlink tab (same models: antenna pattern, atmospheric attenuation and G/T at each elevation, losses of the Static LB · Downlink tab). Enter the **Start Date** and the number of **Days**, select one or more **Ground Stations** (about a second per day and station) and press **Compute Statistics**.

- **Station network**: the satellite is in contact when any selected station sees it (above the minimum elevation and that station's horizon mask); at each instant the station with the best margin is used. Each station uses its own antenna diameter (G/T scaling) and mask; the pass table gives the stations of each contact.
- **Usable minutes**: samples in contact whose margin, Eb/No minus the **Required Eb/No**, is at least the **Required Margin**; when these fields are empty the values of the Static LB · Downlink tab are used.
- **Capacity**: the information bit rate (coded bit rate / overhead) over the usable minutes, in MB of 10^6 bytes: what the link could download.
- **On-board data**: the rates in kbps of what is sent on this downlink band, **Instrument Data**, **HKTM** and **Other Data** (the uplink, telecommands, is not included). With them the on-board memory is simulated, starting empty: it fills at the total rate and empties at the downlink rate during the usable minutes. This gives the volume actually **downloaded**, the data **on board** at the end of each day and its maximum, and the **data latency** (time from generation to arrival on ground, first in first out; mean and maximum). The downlink keeps up when the mean

capacity is at least the data generated per day. The rates are read from and saved to the parameters file (downlink_data_kbps).

- **Latency Requirement** (min) and **Within** (% of data), with the on-board data rates: e.g. 60 min for 90 % of the data. The summary gives the share of the downloaded data on ground within the required time, the time within which the required share arrives, and whether the requirement is met. Saved in the parameters file (latency_requirement).
- **Real-time TM** (kbps): telemetry sent live during the passes and not stored (e.g. real-time HKTM); it takes that part of the downlink bit rate, leaving less for the stored data.
- **On-board Memory** (GB, with the on-board data rates): memory of each satellite for these data, e.g. 4 GB for one partition. The summary gives the largest share used, the hours without contacts until it is full and, if it fills, the data lost (the new data are discarded; the latency of the rest is then approximate). Saved in the parameters file (memory_gb); lbtool.py --memory-gb.
- **Latency at end of pass**: the data downloaded during a pass count as on ground at the end of that pass, when the dump is complete and a product can be processed (the convention of product timeliness requirements); unchecked, each bit counts when it comes down, a few minutes earlier. Saved in the parameters file (latency_at_pass_end); lbtool.py --latency-at-pass-end.
- **Save to Mission** writes these inputs (data rates, latency requirement, memory, minimum pass, antenna setup, latency at end of pass, ground stations) into the parameters.json of the selected mission, keeping the rest of the file; they are loaded again whenever the mission is selected. **Reload** reads them back; the line below the buttons names the file.
- **Min Pass Duration** (min): contacts (usable minutes of one satellite over one station, across midnight) shorter than this are not used, being too short to schedule (acquisition, lock, start of the dump). Empty: 0, every contact is used. Saved in the parameters file (min_pass_min); lbtool.py --min-pass.
- **Antenna Setup** (min, whole constellation): time to configure an antenna for another satellite. After serving a satellite the antenna cannot take a different one for this long (the same satellite resumes at once); a satellite waiting for it counts as a conflict. Saved in the parameters file (antenna_setup_min; 0 by default).
- **Gaps**: time from the end of a contact to the start of the next one; a gap counts for the day the next contact starts.
- For a satellite without link budget parameters only contacts, contact minutes and gaps are given.
- **Whole constellation** (custom-orbit missions with several satellites): all the satellites of the planes (LTAN list) × **Sats/plane**, evenly spaced in each plane and shifted by **Phasing** from one plane to the next (custom orbit fields of the Orbit & Passes tab, saved in the orbit section). Each satellite has its own memory with the same data rates. Each station tracks one satellite at a time and keeps it while the link closes; free stations go to the satellites with more data on board. A satellite ready to download whose stations are all busy waits (**conflict**). The summary gives the system figures (served minutes, conflicts, volume, memory, latency and requirement), the Per Satellite table the figures of each satellite, the Per Station table the use of each station (served minutes, volume, busy time, minutes with more than one satellite usable).
- The **Per Station** table is also given for one satellite and several stations. Normally one antenna per site is used, the others being backups; with **Use backup antennas** (lbtool.py: --backup-antennas) a station serves as many satellites at a time as its antennas (last column of ground_stations.csv). Busy is the share of the antenna time in use, Contention the minutes with more satellites usable than antennas.

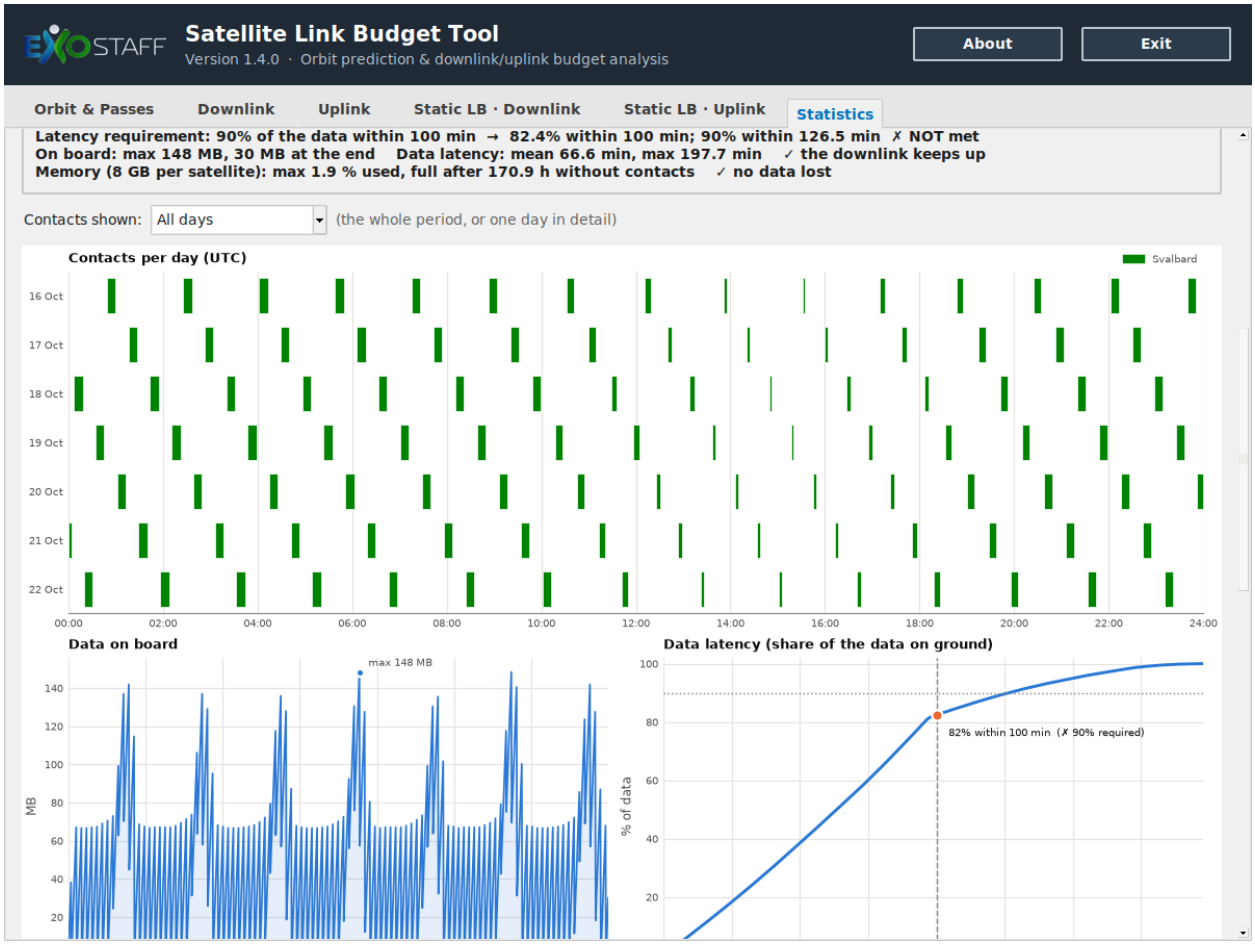


Figure 5 – Statistics plots: one satellite over one station, 104 kbps on board, latency requirement 90 % within 100 min.

Three plots follow the summary. **Contacts**: a timeline, one row per day (for a constellation, one row per satellite over the whole period; **Contacts shown** above the plots draws one day in detail), each pass drawn from its start to its end in UTC and coloured by its station (grey: no usable minute); the gaps are the white spaces. **Data on board**: the memory occupation along the whole period (all satellites for a constellation), with its maximum. **Data latency**: the share of the data on ground within a given latency, with the requirement and whether it is met. Without on-board data rates these two plots become the **gaps between contacts** of each satellite (its revisit time) and the **contact time by station** of each satellite (or day), with the minutes spent waiting for a busy station.

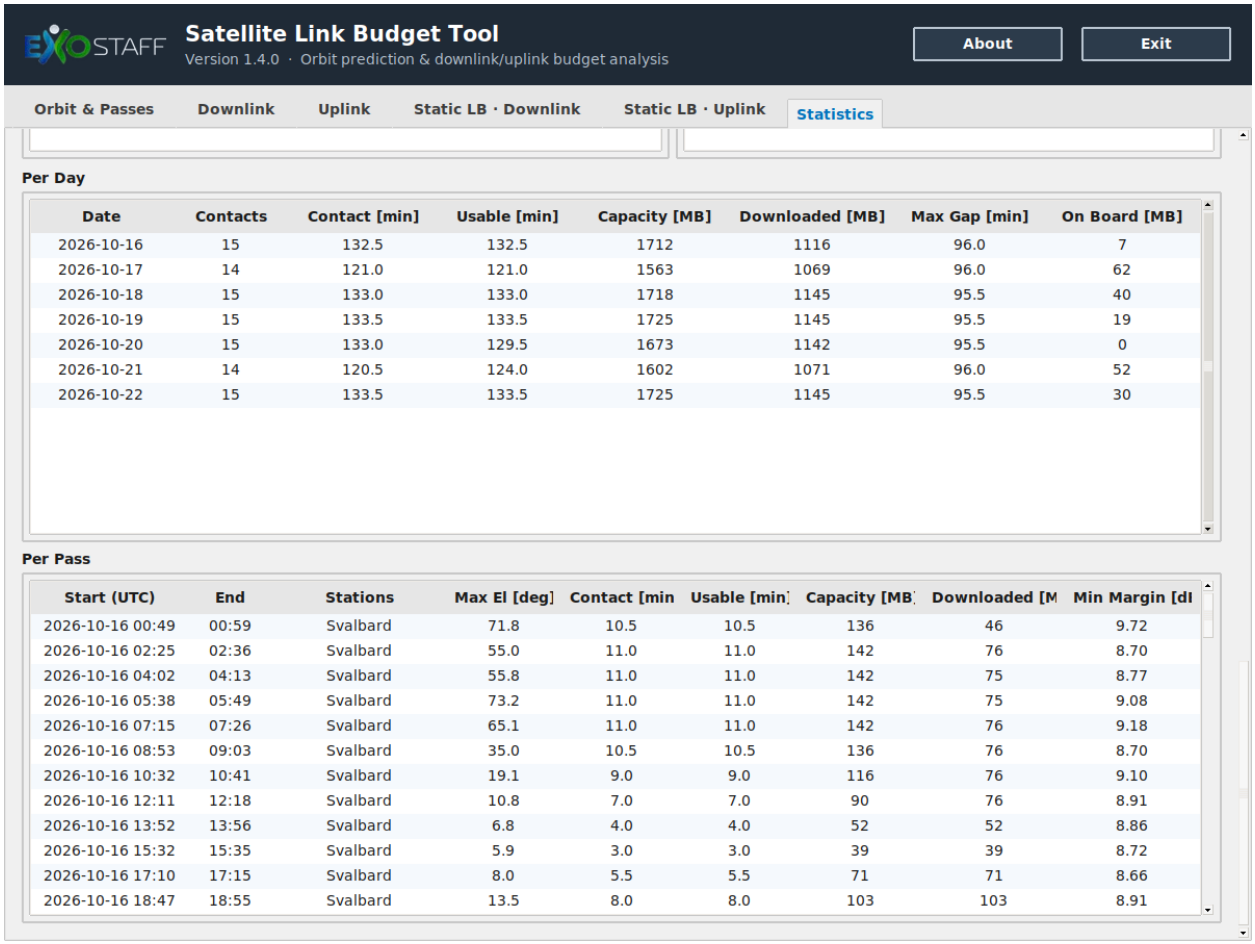


Figure 6 – Per-day and per-pass tables.

The tables give each day (contacts, contact and usable minutes, capacity, downloaded volume, longest gap, data on board) and each pass (times, stations, maximum elevation, minutes, capacity, downloaded volume, minimum margin; a pass with no usable minute is shown in orange). **Export CSV…** writes the summary and both tables. **Export PDF…** writes an A4 report: the verdict on the latency requirement, the inputs used, the summary, the plots and the tables per station, satellite, day and pass (the first 400 passes; all of them are in the CSV).

6 Downlink tab

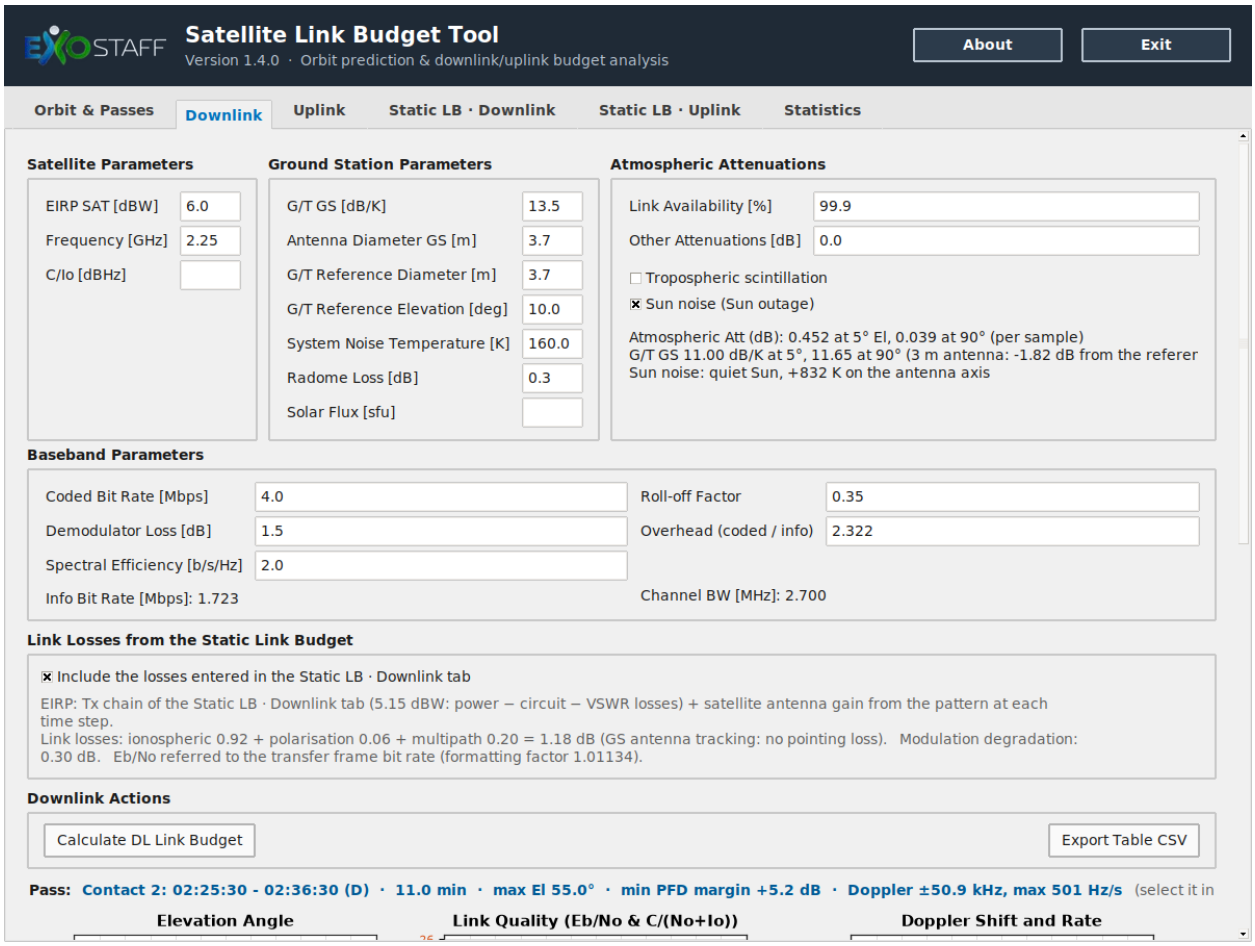


Figure 7 – Downlink tab: parameters and link losses.

6.1 Parameters

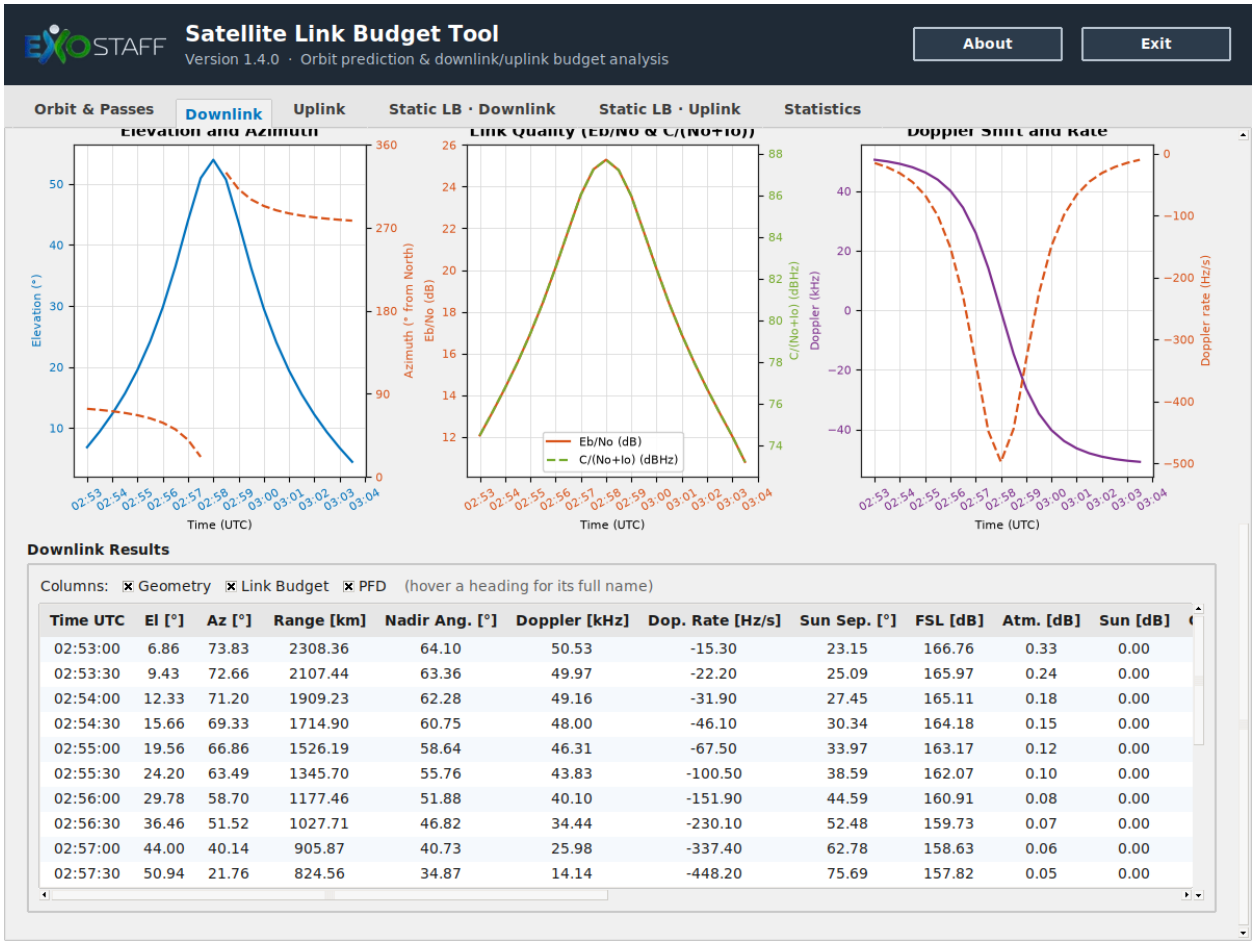
Field	Meaning
EIRP SAT [dBW]	Satellite EIRP. Used by the pass analysis only when the Static LB · Downlink tab uses an EIRP value (not the Tx chain): it is then taken at the antenna pattern peak.
Frequency [GHz]	Downlink carrier frequency.
C/Io [dBHz]	Carrier to interference density. Leave empty when there is no interferer.
G/T GS [dB/K]	Ground station figure of merit of the reference antenna, at the reference elevation in clear sky (e.g. 9.9 dB/K for a 3 m antenna at L band).
Antenna Diameter GS [m]	Used for a station without a diameter in ground_stations.csv (5th column): ITU-R attenuation model, 3 dB beamwidth and G/T scaling.
G/T Reference Diameter [m]	Diameter of the antenna the G/T GS refers to. A station with another diameter gets G/T + 20 log10(D / Dref) (same feed and receiver: a 2.4 m antenna is 1.94 dB below a 3 m one). Empty: no scaling.
G/T Reference Elevation [deg], System Noise Temperature [K]	With a system noise temperature (at the reference elevation, clear sky) the G/T follows the sky noise: T_sky = 275 K × (1 − 10^(−A/10)) + 2.7 K × 10^(−A/10) from the ITU-R attenuation A of the actual elevation and weather (ITU-R P.618), and G/T = G/T_ref − 10 log10((Tsys − T_sky,ref + T_sky) / Tsys). At low elevation and in rain the G/T is lower. Ground spill-over is not modelled. Empty: constant G/T.
Radome Loss [dB]	G/T degradation of the radome (e.g. 0.25 dB), subtracted from the G/T.

Field	Meaning
Solar Flux [sfu]	Solar flux density at the downlink frequency, for the Sun noise (1 sfu = 1e-22 W/m²/Hz). Typically 100–250 sfu at L/S band at solar maximum. Empty: quiet Sun, T = 120000 × f^−0.75 K (about 80 000 K, 49 sfu, at 1.7 GHz).
Link Availability [%]	Rain and clouds are taken for p = 100 − availability [% of time].
Other Attenuations [dB]	Any further fixed loss.
Tropospheric scintillation	ITU-R P.618 tropospheric scintillation; off by default (valid from 5° of elevation and about 4–20 GHz).
Sun noise (Sun outage)	On by default, with a System Noise Temperature. When the Sun is near the ground antenna beam it adds T_sun × Ω_sun × G(θ) / 4π to the system noise temperature (θ: angle Sun–satellite seen from the station; Gaussian main beam of 70 λ/D, peak gain G/T + 10 log10 Tsys; ITU-R S.465 side lobes 32 − 25 log10 θ outside it). G/T and Eb/No drop by 10 log10(1 + ΔT/Tsys). Sun position: Astronomical Almanac low-precision formulae (about 0.01°). A 3 m antenna at 1.7 GHz with the quiet Sun on its axis: +510 K, about −6 dB with Tsys 150 K. The contact table marks the passes with at least 0.5 dB (row in orange; hover it for the noise and the time).
Coded Bit Rate [Mbps]	Channel bit rate of the transmitter (coding included).
Overhead (coded / info)	Coded bit rate / information bit rate (coding × formatting).
Demodulator Loss [dB]	Receiver implementation loss.
Roll-off Factor, Spectral Efficiency	Give the occupied bandwidth: coded rate × (1 + roll-off) / spectral efficiency (QPSK: 2 b/s/Hz).

6.2 Link losses from the static link budget

With **Include the losses entered in the Static LB · Downlink tab** (on by default) the pass analysis uses, from that tab: the Tx chain (transmitter power − circuit and VSWR losses) to which the antenna pattern gain is added, the ionospheric, polarisation and multipath losses, the modulation degradation, and the formatting factor, so that Eb/No is referred to the transfer frame bit rate as in the static budget. The panel shows the values applied. The ground station antenna tracks the satellite and has no pointing loss.

6.3 Results of the selected pass



Time UTC	El [°]	Az [°]	Range [km]	Nadir Ang. [°]	Doppler [kHz]	Dop. Rate [Hz/s]	Sun Sep. [°]	FSL [dB]	Atm. [dB]	Sun [dB]
02:53:00	6.86	73.83	2308.36	64.10	50.53	-15.30	23.15	166.76	0.33	0.00
02:53:30	9.43	72.66	2107.44	63.36	49.97	-22.20	25.09	165.97	0.24	0.00
02:54:00	12.33	71.20	1909.23	62.28	49.16	-31.90	27.45	165.11	0.18	0.00
02:54:30	15.66	69.33	1714.90	60.75	48.00	-46.10	30.34	164.18	0.15	0.00
02:55:00	19.56	66.86	1526.19	58.64	46.31	-67.50	33.97	163.17	0.12	0.00
02:55:30	24.20	63.49	1345.70	55.76	43.83	-100.50	38.59	162.07	0.10	0.00
02:56:00	29.78	58.70	1177.46	51.88	40.10	-151.90	44.59	160.91	0.08	0.00
02:56:30	36.46	51.52	1027.71	46.82	34.44	-230.10	52.48	159.73	0.07	0.00
02:57:00	44.00	40.14	905.87	40.73	25.98	-337.40	62.78	158.63	0.06	0.00
02:57:30	50.94	21.76	824.56	34.87	14.14	-448.20	75.69	157.82	0.05	0.00

Figure 8 – Plots and table of a pass.

The plots show elevation and azimuth (dashed, right axis, degrees from North clockwise; the line breaks where the satellite crosses North), Eb/No and C/(No+Io), and Doppler shift and rate along the pass; the table gives every 30 s step. **Calculate DL Link Budget** recomputes the budget with new parameters without repeating the orbit propagation; **Export Table CSV** saves the table (with project and pass on the first lines). The table uses short headings (hover a heading for its full name and unit) and the **Columns** switches above it show or hide the Geometry, Link Budget and PFD groups; samples above the PFD limit are in red. Main columns (full names, as in the CSV export):

Column	Meaning
Off Boresight Angle (°)	Angle from the satellite nadir to the station.
S/C Antenna Gain (dBi)	Satellite antenna gain at that angle, from the pattern.
EIRP (dBW)	Tx chain + S/C antenna gain (or EIRP SAT at the pattern peak − roll-off).
Atmospheric Att (dB)	ITU-R attenuation at the elevation of the sample (the model is evaluated on an elevation grid and interpolated).
G/T GS (dB/K)	Ground station G/T at the elevation of the sample, with the Sun noise (see the parameters).
Sun Separation (°)	Angle between the Sun and the satellite seen from the station.
Doppler Rate (Hz/s)	Rate of change of the downlink Doppler shift, −f r''/c, from the range acceleration (relative velocity, satellite gravity, station rotation): exact at every sample, peak at the closest approach. Shown dashed in the Doppler plot; the contact table gives the largest Doppler shift and rate of each pass (Doppler column).
Sun Noise (dB)	G/T degradation from the Sun in the antenna beam.

Column	Meaning
Link Losses (dB)	Ionospheric + polarisation + multipath, from the static tab.
C/(No+Io) (dBHz)	Carrier to noise (and interference) density after the demodulator losses.
Eb/No (dB)	C/(No+Io) − 10·log10(transfer frame bit rate).

7 Uplink tab

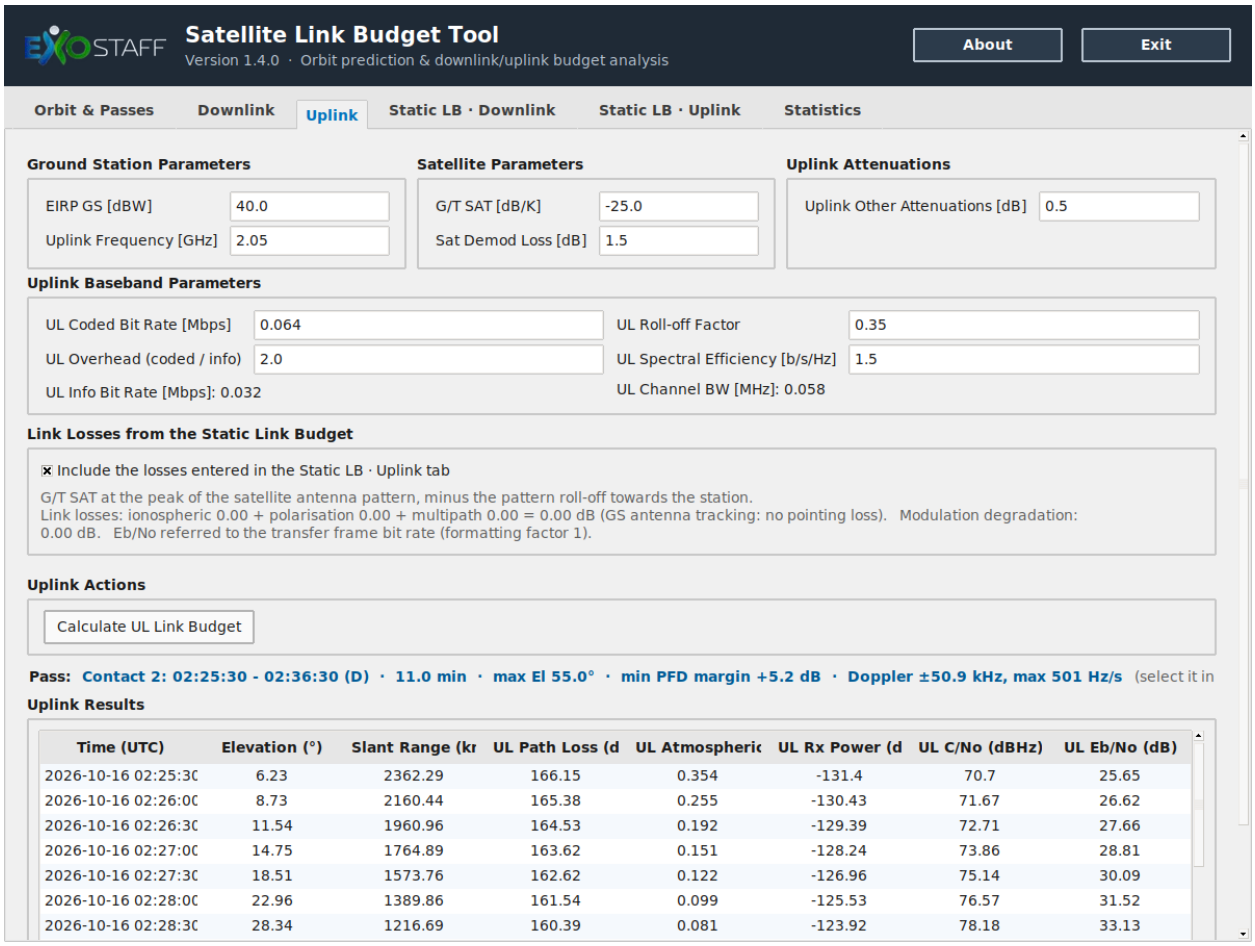


Figure 9 – Uplink tab.

Same principle as the Downlink tab, for the ground station to satellite link: EIRP GS, uplink frequency, G/T SAT, satellite demodulator loss, uplink attenuations and baseband parameters. G/T SAT is taken at the satellite antenna boresight; at each time step the pattern roll-off towards the station (peak gain − gain) is subtracted. **Calculate UL Link Budget** shows the table of the pass selected in the Orbit & Passes tab.

8 Static link budget tabs

8.1 Layout

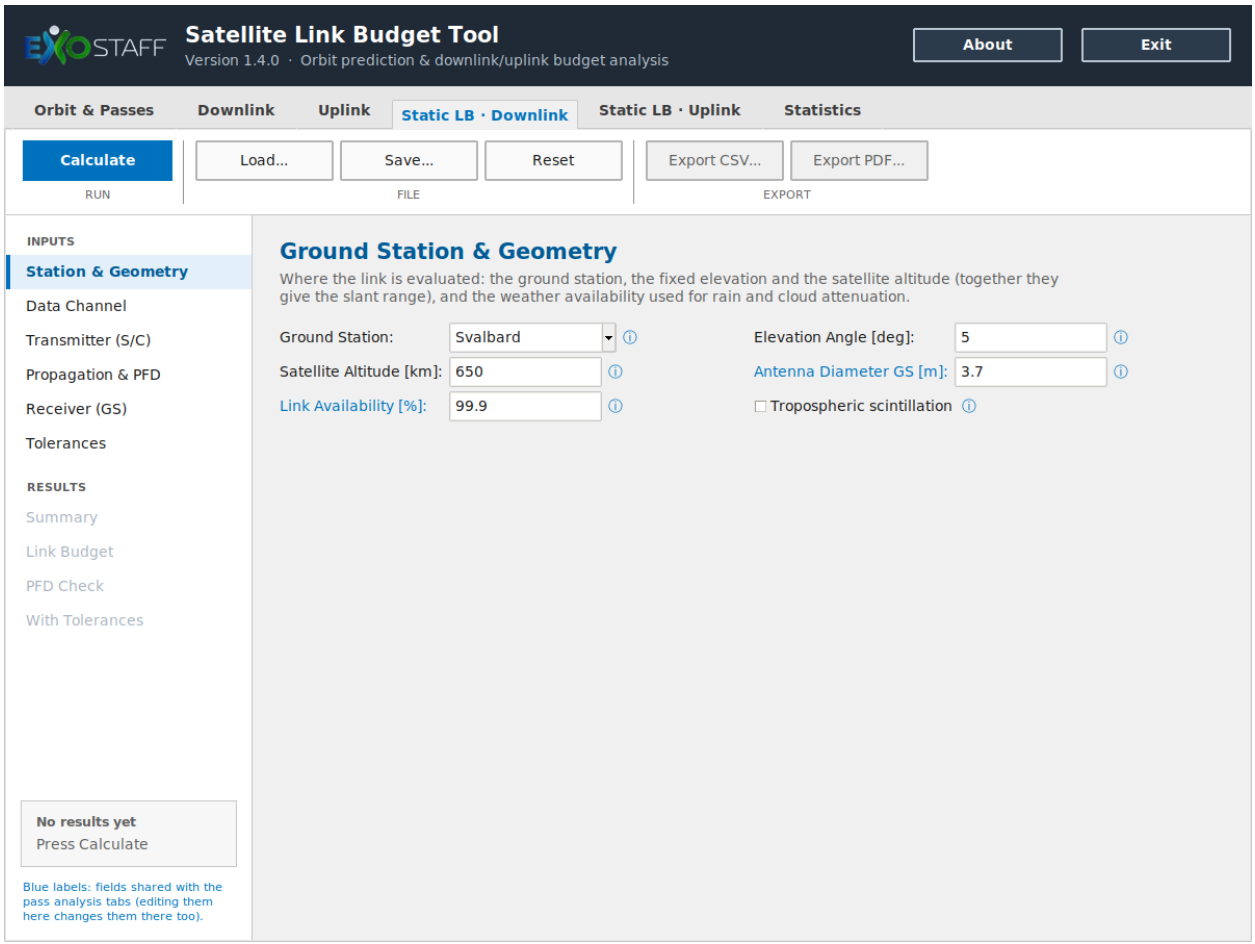


Figure 10 – Static LB · Downlink: toolstrip, sidebar and an input page.

- **Toolstrip**: RUN – **Calculate** (blue when there are no results yet or the inputs changed); FILE – **Load…**, **Save…**, **Reset**; EXPORT – **Export CSV…**, **Export PDF…** (enabled once results are up to date).
- **Sidebar**: the input pages (INPUTS) and the result pages (RESULTS, available after Calculate). At the bottom, the verdict and margin of the last calculation, or OUTDATED when an input changed.
- An input error opens the page of the field concerned and puts the cursor in it.
- **Save…** writes the inputs of the tab into the parameters file (section fixed_elevation_downlink or fixed_elevation_uplink), together with the shared values and the project name, keeping the other keys of the file. Saving into parameters.json makes them load at startup.

8.2 Input pages

Station & Geometry

Ground station (its own selection, independent of the pass analysis; its location sets the ITU-R attenuation and the ionospheric zone), elevation angle (the worst case: the minimum operational elevation) and satellite altitude, which give the slant range; antenna diameter and link availability (shared); tropospheric scintillation (off by default).

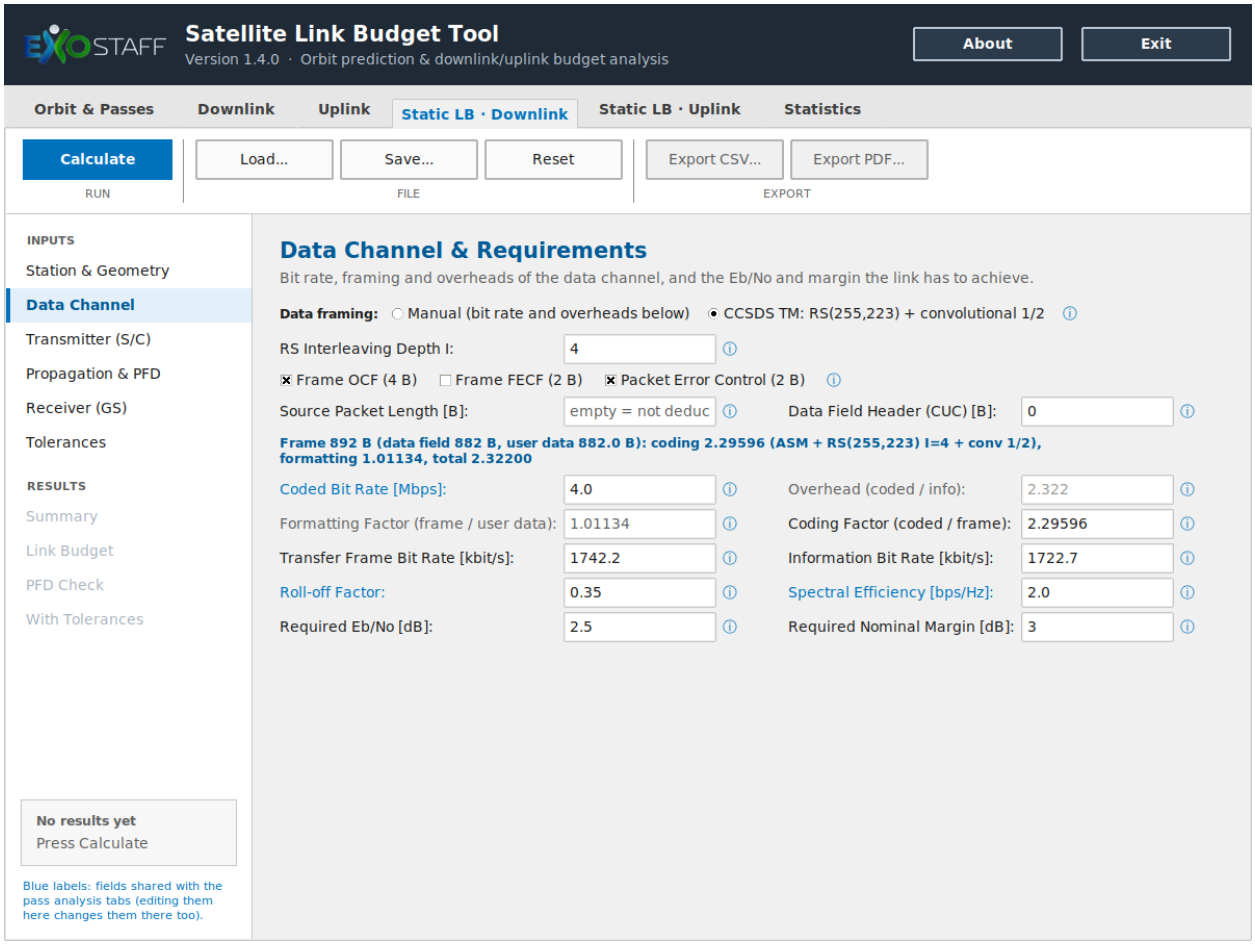


Figure 11 – Data Channel page with the CCSDS framing.

Data Channel & Requirements

Data framing: Manual (enter overhead and formatting factor) or CCSDS TM (downlink): from the coded bit rate and the RS(255,223) interleaving depth I, the tool derives the transfer frame (223·I bytes: 6-byte primary header, optional OCF 4 B and FECF 2 B), the coding factor (ASM 4 B per codeblock of 255·I bytes, convolutional 1/2), the formatting factor (frame / user data; with a Source Packet length also the packet headers, CUC data field header and packet error control are deducted) and the transfer frame and information bit rates, shown in grey boxes. The (i) icons give the detailed breakdown. Then the required Eb/No and required nominal margin.

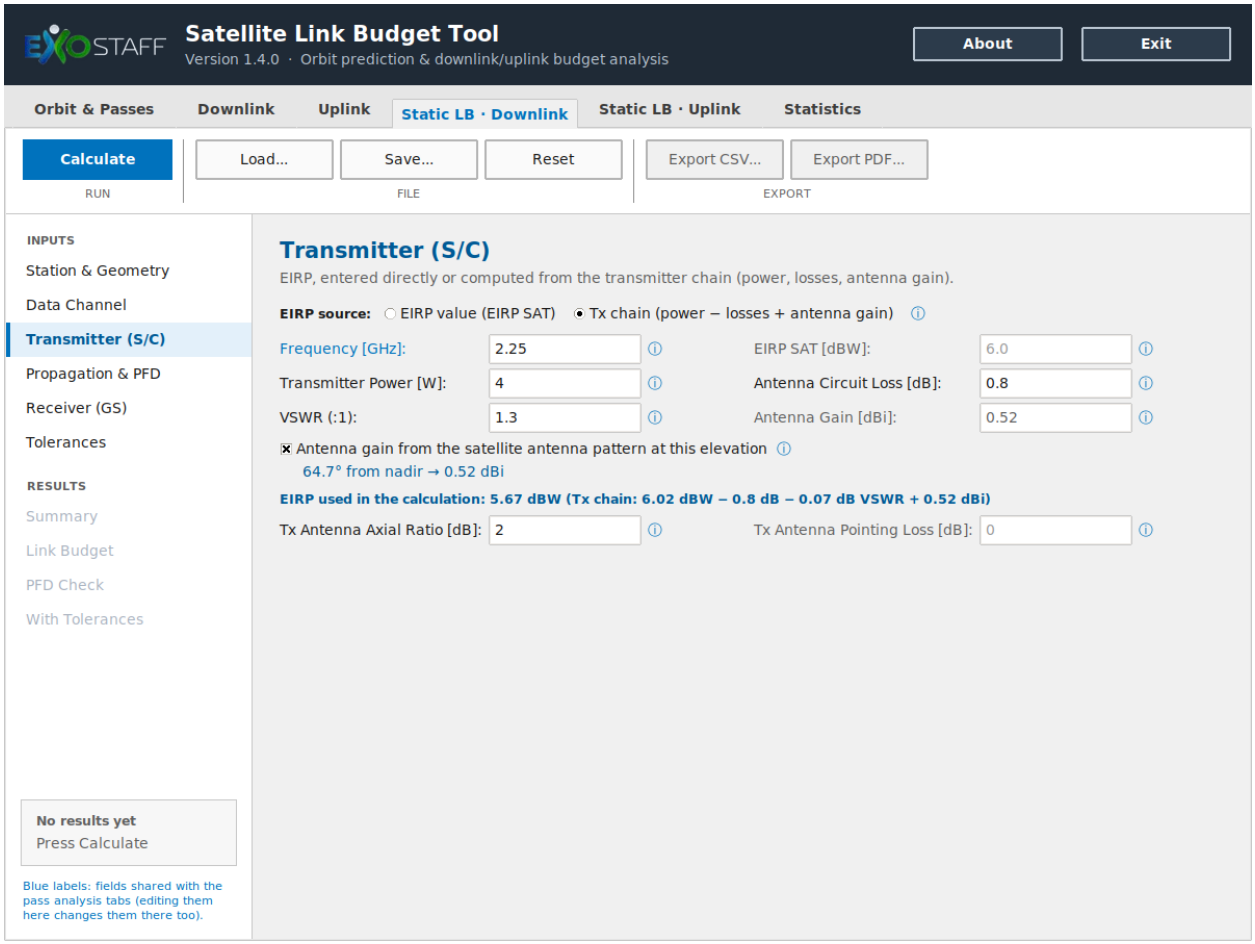


Figure 12 – Transmitter page: Tx chain and antenna gain from the pattern.

Transmitter

EIRP source: EIRP value (shared field) or Tx chain: EIRP = transmitter power − antenna circuit loss − VSWR mismatch loss + antenna gain. On the downlink, **Antenna gain from the satellite antenna pattern at this elevation** reads the gain from the pattern at the angle from nadir of the station (e.g. 65.4° at 3° and 628 km); the Tx antenna pointing loss is then not used. The EIRP actually used is shown under the chain. Tx axial ratio and Tx pointing loss complete the page.

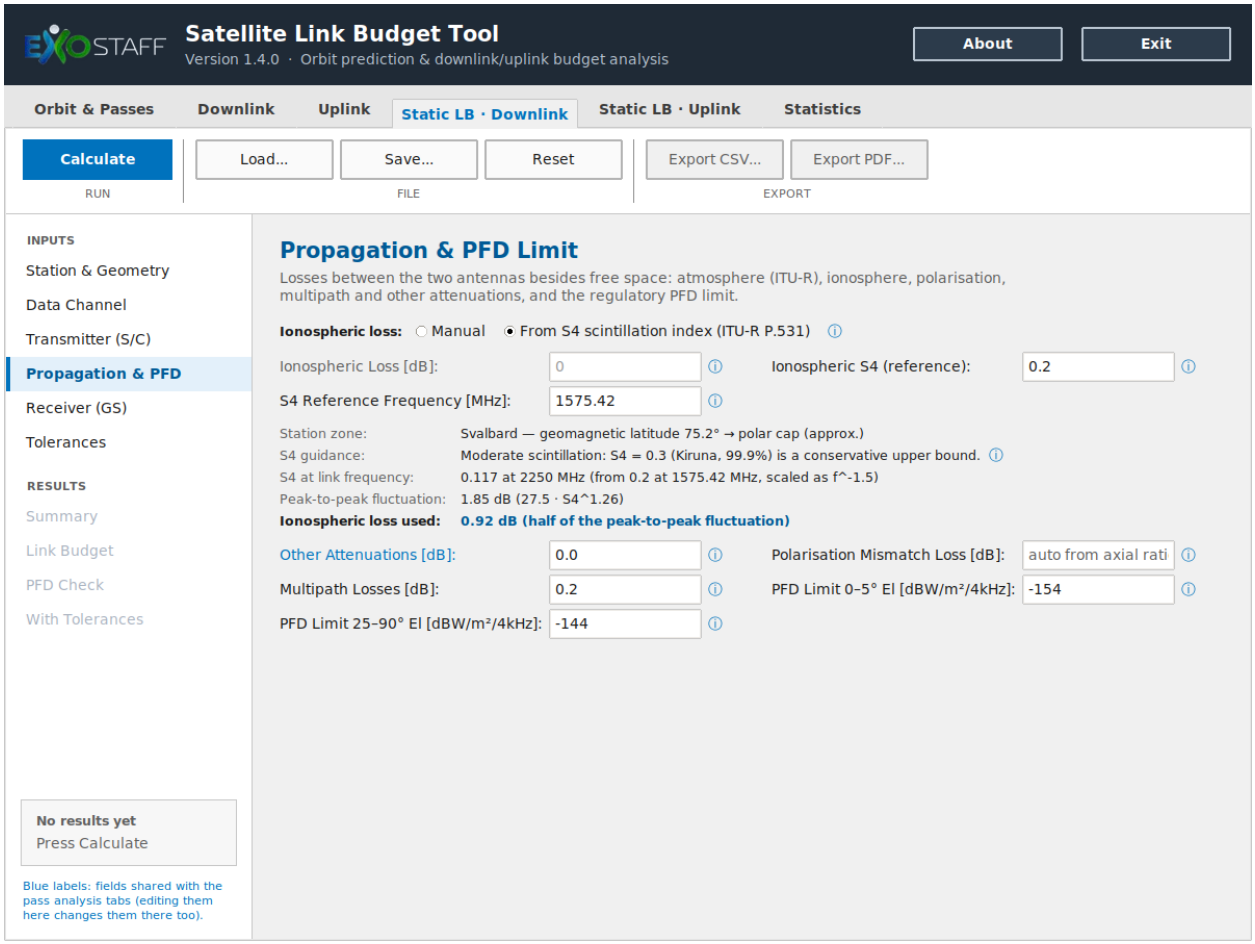


Figure 13 – Propagation page: ionospheric loss from S4.

Propagation & PFD Limit

Ionospheric loss: Manual or From S4 scintillation index (ITU-R P.531, chapter 9.3); the page shows the station's geomagnetic latitude and ionospheric zone as guidance for S4. Then other attenuations (shared; values ≥ 1 dB are flagged, as they may count the atmospheric losses twice), polarisation mismatch loss (entered, or computed from the Tx and Rx axial ratios when empty), multipath losses and, on the downlink, the PFD limits: **PFD Limit 0–5° El** and, optionally, **PFD Limit 25–90° El**, linear in between as in the ITU Radio Regulations Art. 21 (without the second one the first applies at every elevation). The static check uses the limit at its elevation; the pass analysis checks the PFD at every sample against the same limits.

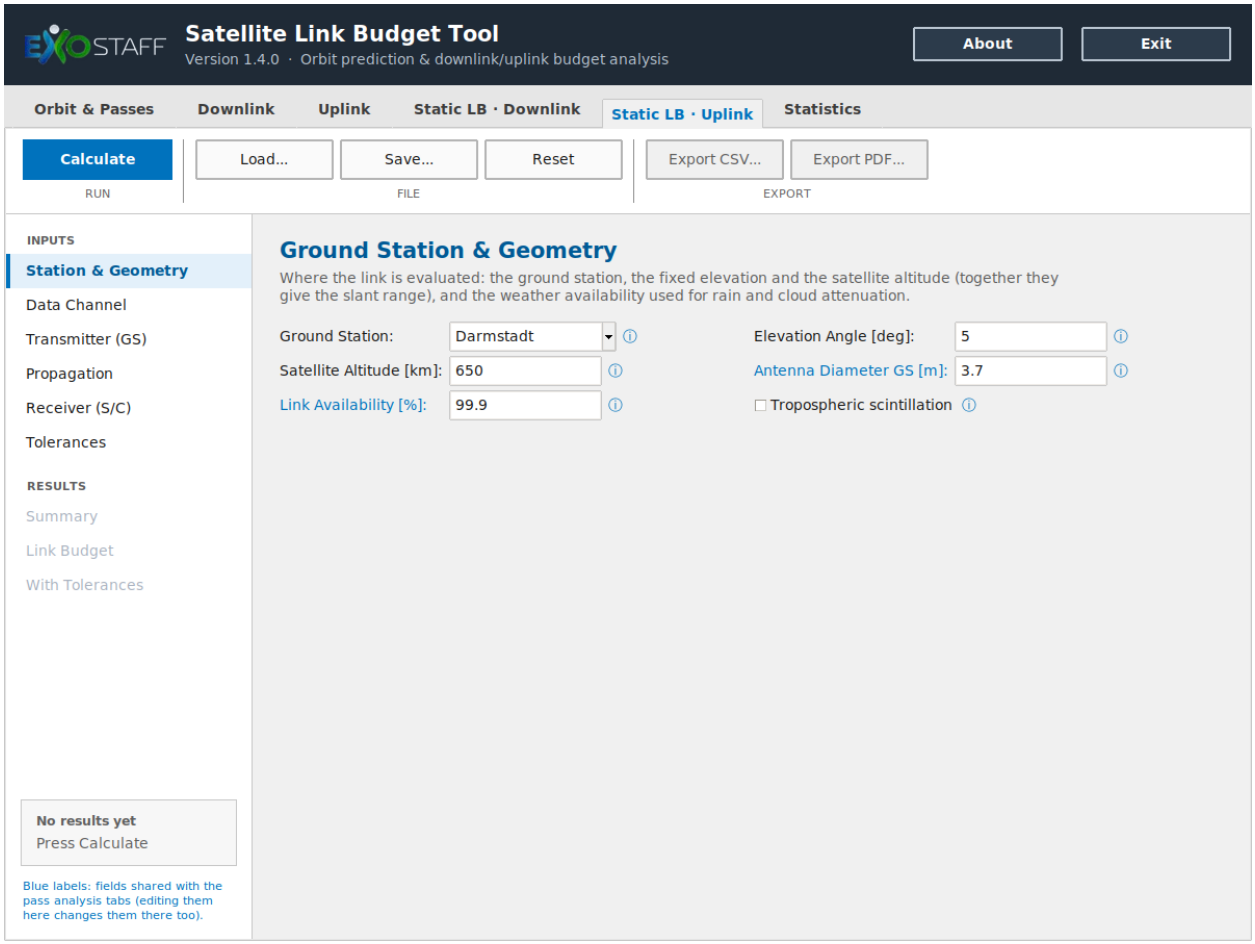


Figure 14 – Static LB · Uplink, Receiver page: G/T SAT from the pattern.

Receiver

G/T and demodulator loss (shared), modulation degradation, Rx axial ratio, Rx antenna depointing angle and 3 dB beamwidth (pointing loss $12 \cdot (\theta/\theta 3\text{dB})^2$; beamwidth $70 \cdot \lambda/D$ from the GS dish when empty). On the uplink, **G/T SAT at the antenna boresight: subtract the pattern roll-off at this elevation** reduces G/T SAT by the pattern roll-off towards the station, as the pass analysis does at each time step; the Rx depointing angle is then not used.

Tolerances

Optional favourable/adverse deviations (positive dB) of transmitter power, Tx antenna gain, propagation loss, Rx G/T and receiver degradation, each with a distribution: TRI (triangular, nominal most likely), UNI (uniform) or GAU (Gaussian with the limits at ±3σ). They give the statistical margins (chapter 9.7).

8.3 Result pages

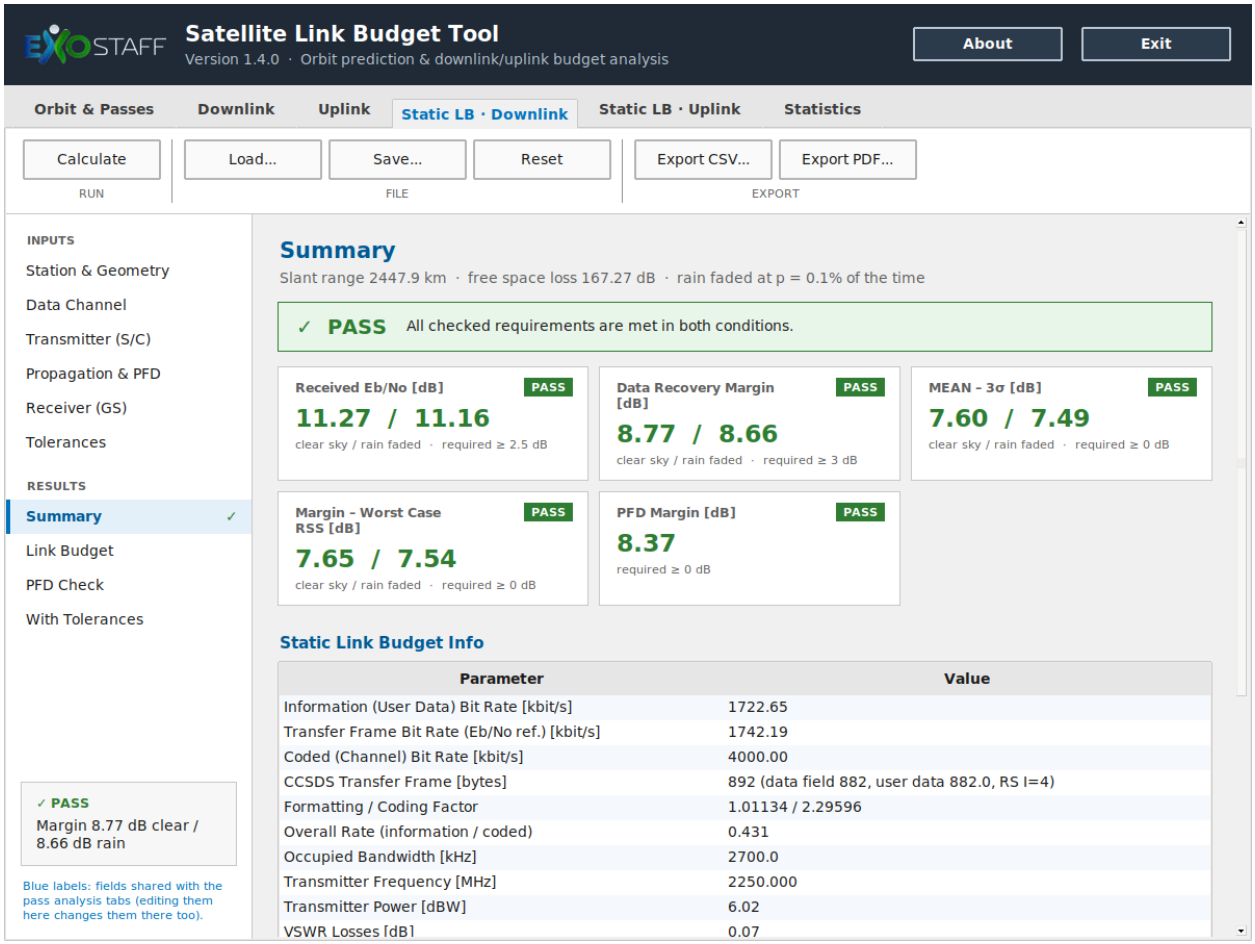


Figure 15 – Summary page.

- **Summary**: overall PASS/FAIL, one tile per key result (Eb/No, Data Recovery Margin, MEAN − 3σ, Worst Case RSS, PFD margin) with clear-sky / rain-faded values and requirement, notes (e.g. elevation below 5°) and the derived parameters.
- **Link Budget**: the full table, grouped as Transmitter, Propagation, Receiver, Demodulation, Margins, for clear sky and rain faded (p = 100 − availability).
- **PFD Check** (downlink): PFD at the ground, in 4 kHz, limit and margin.
- **With Tolerances**: each toleranced quantity down to the margin, with nominal, favourable, adverse, mean, variance and distribution (rain-faded condition).

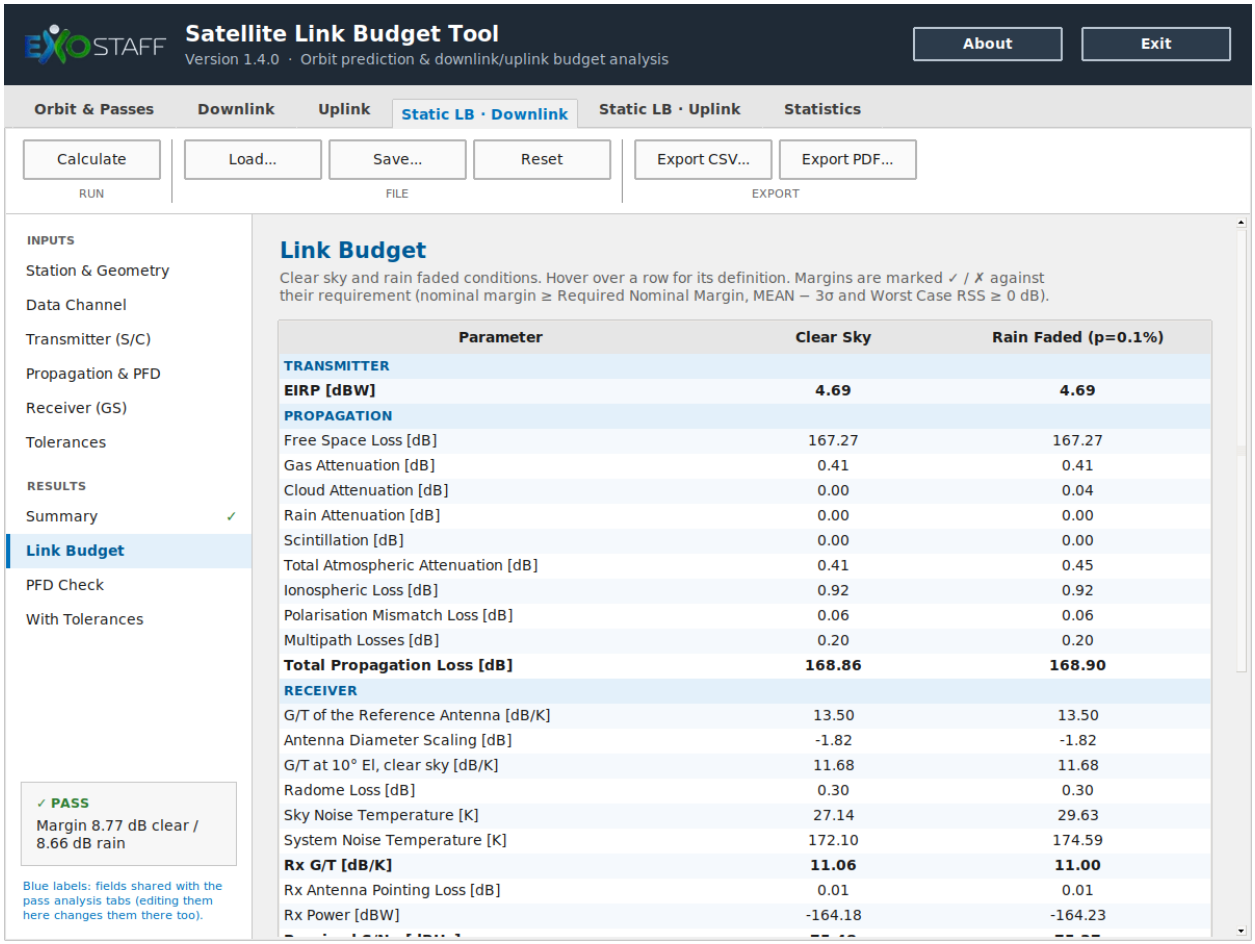


Figure 16 – Link Budget page.

8.4 Exports

Export CSV… saves inputs and results as a table. **Export PDF…** creates an A4 report: summary with PASS/FAIL, input and derived parameters, link budget table, PFD check, link budget with tolerances and the method and assumptions, with the project name and the software version.

9 Calculation method

A summary of the models. Chapter 10 gives, for every parameter and every result, the exact formula and the value used when a field is empty; chapter 11 does the same for the statistics.

9.1 Geometry

Spherical Earth (R = 6378.137 km), circular orbit at altitude h. For an elevation el, the slant range d and the angle η from the satellite nadir to the station are:

$$d = \sqrt{(R+h)^2 - (R+h_{GS})^2 \cos^2 \mathrm{el}} - (R+h_{GS}) \sin \mathrm{el}$$

$$\eta = \arcsin\left[\frac{(R+h_{GS})\cos \mathrm{el}}{R+h}\right]$$

In the pass analysis elevation, range and angle come from the orbit propagation (Skyfield).

9.2 Propagation

$L_{FS} = 20\log_{10} d_{km} + 20\log_{10} f_{GHz} + 92.45 \ \mathrm{dB}$	Free space loss.
$A = A_g + \sqrt{(A_r + A_c)^2 + A_s^2}, \quad p = 100 - \mathrm{availability}$	ITU-R P.618 at the station: gas (P.676), cloud (P.840), rain (P.618/P.837) exceeded p %, optional tropospheric scintillation. Clear sky: gas only.
$L_{pol} = -20\log_{10}\frac{2\,(r_1 + r_2)}{(1+r_1)(1+r_2)}, \quad r = 10^{AR/20}$	Polarisation mismatch of co-rotating elliptical antennas, co-polar value.

9.3 Ionospheric scintillation

From the S4 index measured at a reference frequency (GPS L1 1575.42 MHz by default), as in the ESA method for L-band downlinks:

$S_4(f) = S_{4,\,\mathrm{ref}}\left(\frac{f}{f_{\mathrm{ref}}}\right)^{-1.5}, \quad L_{iono} = \frac{27.5\,S_4^{1.26}}{2} \ \mathrm{dB}$	The fade is half of the peak-to-peak fluctuation.

The geomagnetic latitude (dipole) gives the zone: equatorial (< 20°), mid-latitude (< 58°), auroral oval (< 72°), polar cap.

9.4 Antennas

$\mathrm{EIRP}(\eta) = P_{Tx} - L_{circuit} - L_{VSWR} + G(\eta)$	Downlink: the pattern gives the gain at the angle from nadir η (0 dBi outside the table).
$(G/T)_{SAT}(\eta) = (G/T)_{SAT} - [G_{peak} - G(\eta)]$	Uplink.
$L_{point} = 12\left(\frac{\theta}{\theta_{3dB}}\right)^2, \quad \theta_{3dB} = 70\,\frac{\lambda}{D}$	Ground antenna depointing (static link budget); along the passes it tracks the satellite.

9.5 Carrier to noise and Eb/No

$$P_{\mathrm{Rx}} = \mathrm{EIRP} - L_{\mathrm{FS}} - A - L_{\mathrm{other}} - L_{\mathrm{iono}} - L_{\mathrm{pol}} - L_{\mathrm{mp}} - L_{\mathrm{point}}$$

Received power [dBW].

$$S/N_0 = P_{\mathrm{Rx}} + G/T + 228.6$$

Boltzmann constant: −228.6 dBW/K/Hz.

$$C/N_0 = S/N_0 - L_{\mathrm{demod}} - L_{\mathrm{mod}}$$

Combined with C/I_0 when given (pass analysis).

$$R_{\mathrm{frame}} = \frac{R_c}{OH} F$$

Transfer frame bit rate.

$$E_b/N_0 = C/N_0 - 10\log_{10} R_{\mathrm{frame}}, \quad M = E_b/N_0 - (E_b/N_0)_{\mathrm{req}}$$

M: Data Recovery Margin.

9.6 Power flux density

$$\mathrm{PFD} = \mathrm{EIRP} - 10\log_{10}(4\pi d^2)\ [\mathrm{dBW/m^2}]$$

No atmospheric loss credit (conservative).

$$\mathrm{PFD}_{4\mathrm{kHz}} = \mathrm{PFD} - 10\log_{10}\frac{B}{4\ \mathrm{kHz}}, \quad B = \frac{R_c(1+\alpha)}{\eta_s}$$

Occupied bandwidth B.

$$M_{\mathrm{PFD}} = \mathrm{limit(el)} - \mathrm{PFD}_{4\mathrm{kHz}} \geq 0$$

ITU Radio Regulations Art. 21: the 0–5° limit up to 5°, the 25–90° limit from 25°, linear between.

Along the passes the PFD is computed at every sample with the EIRP of the pattern towards the station; the contact table shows in red a pass that exceeds the limit (hover the row for by how much), and the pass table in red its samples above the limit; the PFD Check page of the static budget gives the limit and the margin.

9.7 Statistical margins

Each tolerance with adverse limit a = −adverse and favourable limit b = +favourable contributes a mean μ and a variance σ²; contributions are independent, so means and variances add:

$$\mathrm{TRI}: \ \mu = \frac{a+b}{3}, \ \sigma^2 = \frac{a^2+b^2-ab}{18} \qquad \mathrm{UNI}: \ \mu = \frac{a+b}{2}, \ \sigma^2 = \frac{(b-a)^2}{12}$$

$$\mathrm{GAU}: \ \mu = \frac{a+b}{2}, \ \sigma^2 = \left(\frac{b-a}{6}\right)^2$$

Limits at ±3σ.

$$M_{-3\sigma} = M_{\mathrm{nom}} + \sum_k \mu_k - 3\sqrt{\sum_k \sigma_k^2}, \quad M_{\mathrm{RSS}} = M_{\mathrm{nom}} - \sqrt{\sum_k \mathrm{adv}_k^2}$$

MEAN − 3σ and Worst Case RSS.

10 Parameter and calculation reference

This chapter documents every input of the program and every value it computes: where the input is entered, its key in **parameters.json**, its unit, what the program assumes when it is left empty, and the exact formula through which it reaches the results. It is written from the source code; the Python function that implements each step is named in italics (module.function), so that every statement can be checked against the program. Chapter 9 is the short summary of the same methods.

Notation: log is the base-10 logarithm; quantities in dB, dBW, dBi, dB/K, dBHz add and subtract; $x_{lin} = 10^{x/10}$ converts a dB value to a linear ratio. Angles are in degrees unless written in radians. 'Empty' means a field left blank or a null value in parameters.json.

10.1 Constants and units

Quantity	Value	Used for
Earth radius R	6378.137 km (WGS-84 equatorial)	Static geometry, synthetic TLE (altitude = a − R).
Earth gravitational parameter μ	398600.4418 km³/s²	Synthetic TLE, Doppler rate.
J2	$1.08262668 \cdot 10^{-3}$	Sun-synchronous inclination, nodal period.
Sun-synchronous node rate	2π / (365.2421897 · 86400) rad/s	Sun-synchronous inclination.
Earth rotation rate	$7.2921159 \cdot 10^{-5}$ rad/s	Doppler rate (station acceleration).
Speed of light c	299 792.458 km/s	Doppler, wavelength λ = c / f.
Boltzmann constant k	$1.380649 \cdot 10^{-23}$ J/K; −10 log k = 228.6 dBW/K/Hz	C/No = Rx power + G/T + 228.6.
Mean radiating temperature Tm	275 K (ITU-R P.618 §3)	Sky noise temperature.
Cosmic background Tc	2.7 K	Sky noise temperature.
Sun angular radius	0.2665° → solid angle $\Omega_{sun} = \pi (0.2665° \text{ in rad})^2 = 6.80 \cdot 10^{-5}$ sr	Sun noise.
Geomagnetic north pole	80.65° N, 72.68° W (IGRF-13, epoch 2020)	Geomagnetic latitude, ionospheric zone.
GPS L1	1575.42 MHz	Default reference frequency of S4.
Time step	30 s	Pass analysis and statistics; days start at 00:00 UTC.
Data units	kbps = 10^3 bit/s, Mbps = 10^6 bit/s; MB = 10^6 bytes = $8 \cdot 10^6$ bit; GB = 10^9 bytes	On-board memory in GB (memory_gb × $8 \cdot 10^9$ bit), volumes in MB.
Atmospheric models	ITU-R P.618-13 and the recommendations it calls (P.676 gas, P.840 cloud, P.837/P.838/P.839 rain), library itur 0.4	Atmospheric attenuation.

10.2 List of symbols

Every symbol used in the formulas of chapters 9, 10 and 11, with its meaning, its unit and the section where it is defined or computed. Subscripts: GS ground station, SAT satellite, UL uplink, ref reference, req required.

Orbit and geometry

Symbol	Meaning	Unit	Section
R	Earth equatorial radius, 6378.137	km	10.1
$\mu,\ J_2$	Earth gravitational parameter, second zonal harmonic	km³/s², –	10.1
$a,\ e,\ i,\ \omega$	Semi-major axis, eccentricity, inclination, argument of perigee	km, –, °, °	10.3
h	Satellite altitude (a − R)	km	10.3
$n,\ p$	Mean motion, semi-latus rectum a(1 − e²)	rad/s, km	10.3
$\Omega,\ \dot{\Omega}_{\mathrm{SSO}}$	Right ascension of the ascending node (RAAN); node rate of a sun-synchronous orbit	°, rad/s	10.3
$M_0,\ \dot{M},\ \dot{\omega}$	Mean anomaly at the epoch; J2 rates of mean anomaly and perigee	°, rad/s	10.3
T	Nodal period	min	10.3
$u,\ \Delta u$	Argument of latitude at the epoch; phasing between planes	°	10.3
$N,\ p,\ j$	Satellites per plane; plane index; satellite index in the plane	–	10.3
$t_0,\ \mathrm{LTAN}$	Epoch (00:00 UTC of the day); local time of the ascending node	–, h	10.3
$\mathrm{el},\ \mathrm{az}$	Elevation and azimuth of the satellite seen from the station	°	10.4
$\mathrm{el}_{\min},\ \mathrm{el}_{\mathrm{req}},\ m$	Min Elevation; required elevation; horizon mask	°	10.4
h_{GS}	Station altitude	km	10.5
d	Slant range	km (m in the PFD)	10.5
η	Angle from the satellite nadir to the station (off-boresight angle)	°	10.5
$f_D,\ \dot{f}_D$	Doppler shift and Doppler rate	Hz, Hz/s	10.5
$\dot{r},\ \ddot{r}$	Range rate and range acceleration	km/s, km/s²	10.5
$\mathbf{v}_{\mathrm{rel}},\ \hat{\mathbf{r}}$	Relative velocity; unit vector from the station to the satellite	km/s, –	10.5
$\mathbf{a}_{\mathrm{sat}},\ \mathbf{a}_{\mathrm{GS}}$	Gravity acceleration of the satellite; centripetal acceleration of the station	km/s²	10.5
θ	Angle between two directions (Sun separation, depointing)	°	10.5, 10.8

Propagation and losses

Symbol	Meaning	Unit	Section
$f,\ \lambda$	Carrier frequency; wavelength c / f	GHz, m	10.9
p	Percentage of time the attenuation is exceeded: 100 − availability	%	10.6

Symbol	Meaning	Unit	Section
$A,\ A_g,\ A_c,\ A_r,\ A_s$	Total, gas, cloud, rain, tropospheric scintillation attenuation	dB	10.6
L_{FS}	Free space loss	dB	10.11
$L_{\mathrm{iono}},\ S_4,\ P_{\mathrm{pp}}$	Ionospheric loss; S4 scintillation index; peak-to-peak fluctuation	dB, –, dB	10.14
$f_{\mathrm{ref}},\ S_{4,\mathrm{ref}}$	Frequency at which S4 was measured; S4 at that frequency	MHz, –	10.14
$\phi,\ \lambda,\ \phi_m,\ \phi_p,\ \lambda_p$	Station latitude and longitude; geomagnetic latitude; geomagnetic pole	°	10.14
$L_{\mathrm{pol}},\ AR,\ r$	Polarisation mismatch loss; axial ratio; voltage axial ratio $10^{\mathrm{AR}/20}$	dB, dB, –	10.14
$L_{\mathrm{mp}},\ L_{\mathrm{other}}$	Multipath losses; Other Attenuations	dB	10.12
L_{link}	Static losses applied along the passes: $L_{\mathrm{iono}} + L_{\mathrm{pol}} + L_{\mathrm{mp}}$	dB	10.11
L_{prop}	Total propagation loss of the static budget	dB	10.13
$L_{\mathrm{point}},\ \theta_p,\ \theta_{3\mathrm{dB}}$	Pointing loss; depointing angle; 3 dB beamwidth 70 λ/D	dB, °, °	10.13

Antennas, G/T and Sun

Symbol	Meaning	Unit	Section
P_{Tx}	Transmitter power (10 log of the power in W)	dBW	10.13
$L_{\mathrm{circuit}},\ L_{\mathrm{VSWR}},\ \Gamma$	Antenna circuit loss; mismatch loss; reflection coefficient	dB, dB, –	10.13
T_{chain}	Tx chain without antenna gain: $P_{\mathrm{Tx}} - L_{\mathrm{circuit}} - L_{\mathrm{VSWR}}$	dBW	10.11
$G_{\mathrm{ant}},\ G_{\mathrm{eff}}$	Tx antenna gain; effective gain $G_{\mathrm{ant}} - L_{\mathrm{circuit}} - L_{\mathrm{VSWR}}$	dBi	10.13
$G(\eta),\ G_{\mathrm{peak}}$	Satellite antenna pattern gain at η; highest gain of the pattern	dBi	10.11
EIRP	Effective isotropic radiated power	dBW	10.11
$D,\ D_{\mathrm{ref}}$	Ground antenna diameter; diameter the G/T GS refers to	m	10.7
$(G/T)_{\mathrm{GS}},\ (G/T)_{\mathrm{ref}}$	G/T GS entered; G/T of the station at the reference elevation	dB/K	10.7
$\mathrm{el}_{\mathrm{ref}}$	G/T reference elevation	°	10.7
$T_{\mathrm{sys}},\ T_{\mathrm{sys,ref}}$	System noise temperature; at the reference elevation, clear sky	K	10.7
$T_{\mathrm{sky}},\ T_{\mathrm{sky,ref}}$	Sky noise temperature; at the reference elevation, clear sky	K	10.7
$T_m,\ T_c$	Mean radiating temperature 275 K; cosmic background 2.7 K	K	10.1
L_{radome}	Radome loss	dB	10.7
$T_b,\ S,\ \Omega_\odot$	Sun brightness temperature; solar flux; solar disc solid angle	K, sfu, sr	10.8
$G_{\mathrm{main}},\ G_{\mathrm{side}}$	Ground antenna main-lobe and side-lobe gain towards the Sun	dBi	10.8
$\Delta T,\ \Delta_{\mathrm{Sun}}$	Noise added by the Sun; Sun noise (drop of G/T)	K, dB	10.8

Symbol	Meaning	Unit	Section
k	Boltzmann constant (−228.6 dBW/K/Hz); in 10.3 the J2 coefficient	–	10.1
$n,\ L,\ g,\ \lambda_{\odot},\ \varepsilon$	Days from J2000; Sun mean longitude, mean anomaly, ecliptic longitude; obliquity	d, °	10.8
$\alpha,\ \delta,\ H,\ \mathrm{GMST}$	Sun right ascension, declination, hour angle; sidereal time (α is the roll-off elsewhere)	°	10.8
λ_{GS}	Station longitude	°	10.8

Link budget and data

Symbol	Meaning	Unit	Section
P_{Rx}	Received power	dBW	10.11
$S/N_0,\ C/N_0,\ C/I_0$	Signal and carrier to noise density; carrier to interference density	dBHz	10.11
$L_{\mathrm{demod}},\ L_{\mathrm{mod}},\ D_{\mathrm{mod}}$	Demodulator loss; modulation degradation ($L_{\mathrm{mod}} = D_{\mathrm{mod}}$)	dB	10.9, 10.12
$E_b/N_0,\ (E_b/N_0)_{\mathrm{req}}$	Energy per bit to noise density; required	dB	10.11
$R_c,\ R_{\mathrm{frame}},\ R_{\mathrm{info}}$	Coded, transfer frame and information bit rate	bit/s	10.9, 10.14
$OH,\ F,\ c$	Overhead R_c/R_{info}; formatting factor $R_{\mathrm{frame}}/R_{\mathrm{info}}$; coding factor R_c/R_{frame}	–	10.14
$I,\ L_{\mathrm{frame}},\ L_{\mathrm{cb}},\ L_{\mathrm{df}},\ L_{\mathrm{user}}$	RS interleaving depth; frame, codeblock, data field and user data lengths	–, bytes	10.14
$L_{\mathrm{SH}},\ L_{\mathrm{pkt}},\ L_{\mathrm{CUC}},\ L_{\mathrm{PEC}}$	Frame secondary header; packet length; packet data field header; packet error control	bytes	10.14
$B,\ \alpha,\ \eta_s$	Occupied bandwidth; roll-off; spectral efficiency	Hz, –, b/s/Hz	10.9
$\mathrm{PFD},\ \mathrm{PFD}_{4\mathrm{kHz}}$	Power flux density; in the 4 kHz reference bandwidth	dBW/m², dBW/m²/4kHz	10.13
$L_{\mathrm{low}},\ L_{\mathrm{high}},\ M_{\mathrm{PFD}}$	PFD limits 0–5° and 25–90°; PFD margin	dBW/m²/4kHz, dB	10.14

Margins and tolerances

Symbol	Meaning	Unit	Section
$M,\ M_{\mathrm{nom}}$	Data Recovery Margin $E_b/N_0 - (E_b/N_0)_{\mathrm{req}}$ (nominal)	dB	10.13
$a,\ b,\ \mathrm{adv},\ \mathrm{fav}$	Adverse (negative) and favourable limits of a tolerance	dB	10.14
$\mu_k,\ \sigma_k^2,\ \sigma$	Mean and variance of tolerance k; standard deviation of the margin	dB, dB², dB	10.14
$M_{\mathrm{mean}},\ M_{-3\sigma},\ M_{\mathrm{RSS}}$	Margin mean, MEAN − 3σ, Worst Case RSS	dB	10.14
$M_{\mathrm{fav}},\ M_{\mathrm{adv}}$	Margin with all tolerances favourable / adverse	dB	10.14

Statistics

Symbol	Meaning	Unit	Section
$s,\ g,\ t,\ \Delta t$	Satellite, station, sample; time step 30 s	–, –, –, s	11.2
$m(s,g,t),\ M_{\mathrm{req}}$	Margin over the required Eb/No; Required Margin	dB	11.2

Symbol	Meaning	Unit	Section
$C,\ R_d,\ R_{\mathrm{RT}}$	Capacity of a sample; downlink rate of the stored data; real-time TM rate	bit, bit/s, bit/s	11.2
G	Data generated on board: instruments + HKTM + other	bit/s	11.3
$Q,\ Q_{\max}$	Data on board; on-board memory	bit	11.3
$x,\ D$	Bits downloaded in a sample; bits downloaded before it	bit	11.3
t_{arr}	Time the chunk reaches the ground (end of sample or of pass)	s	11.3
$\ell_{\min},\ \ell_{\max},\ \bar{\ell}$	Latency range of a chunk; mean latency	s, min	11.3
$s(L),\ L_X$	Share of the data within L minutes; latency at X %	–, min	11.3
$P,\ k,\ T_k$	Product length; product number; time the product is complete on ground	min, –, s	11.3
t_{gen}	Time a datum was generated (sensed) on board	s	11.3

10.3 Orbit, TLE and constellation

Launched satellite: the two TLE lines of **tle.txt** in the mission folder, or downloaded by NORAD ID (key `norad_id`; sources SatNOGS, then N2YO). The orbit is propagated with SGP4 (Skyfield) at every 30 s sample. The station is a WGS-84 geodetic point at its latitude, longitude and altitude.

Mission not launched yet (custom orbit, `"custom_orbit": true` in satellites.json): the program writes a synthetic TLE from the `orbit` section of parameters.json (tle_source.synthetic_tle):

Parameter	Key / unit	Empty	Use
Mean altitude	`orbit.altitude_km [km]`	628	Semi-major axis a = R + altitude (100–40000 km).
Eccentricity	`orbit.eccentricity`	0	e, 0 ≤ e < 0.1.
Argument of perigee	`orbit.arg_perigee_deg [deg]`	0	ω (90 for a frozen orbit).
Inclination	`orbit.inclination_deg [deg]`	SSO	i; empty gives the sun-synchronous inclination below.
Planes (LTAN)	`orbit.ltan [hh:mm, list]`	22:20	Local time of the ascending node of each plane; sets the RAAN.
Satellites per plane	`orbit.satellites_per_plane`	1	N satellites evenly spaced in each plane.
Phasing between planes	`orbit.phasing_deg [deg]`	0	Shift of the argument of latitude from one plane to the next.
Extra satellites	`orbit.extra_satellites [LTAN@deg, list]`	none	Single satellites added, e.g. 19:30@120.

The elements of the synthetic TLE are computed as follows:

$$n = \sqrt{\mu / a^3}, \quad p = a\,(1 - e^2)$$

Mean motion and semi-latus rectum.

$$\cos i_{\mathrm{SSO}} = -\frac{\dot{\Omega}_{\mathrm{SSO}}}{1.5\, n\, J_2\, (R/p)^2}$$

Sun-synchronous inclination, when the inclination is empty.

$$\Omega = \alpha_{\odot}(t_0) + 15^{\circ}\,(\mathrm{LTAN} - 12\,\mathrm{h})$$

RAAN; $\alpha_{\odot}$: right ascension of the Sun at the epoch t_0.

$$M_0 = u - \omega$$

Mean anomaly; u: argument of latitude of the satellite at the epoch.

$$k = \frac{3}{4} J_2 \left(\frac{R}{p}\right)^2$$

$$\dot{M} = n\left[1 + k\sqrt{1-e^2}\,(3\cos^2 i - 1)\right], \quad \dot{\omega} = n\,k\,(5\cos^2 i - 1)$$

J2 secular rates of the mean anomaly and of the perigee.

$$T = \frac{2\pi}{\dot{M} + \dot{\omega}}$$

Nodal period.

The epoch is 00:00 UTC of the day analysed (of the first day for the statistics); the drag term is 0. The mean motion written in the TLE is adjusted (up to 5 iterations, to 10^{-10}) until SGP4's own $\dot{M} + \dot{\omega}$ equals $2\pi / T$: SGP4 then flies the J2 nodal period, and the pass times stay right over the analysed days (a Keplerian mean motion would drift by about a minute per orbit). The Sun right ascension uses the low-precision formulae of the Astronomical Almanac (section 10.8).

Constellation (link_core.constellation_members): with P planes in the LTAN list, the satellite j = 1…N of plane p = 0…P−1 has the argument of latitude at the epoch

$$u_{p,j} = (j-1)\,\frac{360^{\circ}}{N} + p\,\Delta u \pmod{360^{\circ}}$$

Δu: phasing between planes.

An extra satellite "LTAN@u" is added with that plane and that u, numbered after the regular satellites of its plane. The satellites are named "<mission> <LTAN> #j" and get the catalogue numbers 99900, 99901, …

Example (an invented mission): 650 km, e = 0.001, inclination empty. The sun-synchronous inclination is 97.986° and the nodal period 97.85 min. With planes 10:30 and 22:30, N = 3 satellites per plane and phasing 60°, u is 0°, 120°, 240° in the 10:30 plane and 60°, 180°, 300° in the 22:30 plane; an extra satellite "10:30@90" becomes "<mission> 10:30 #4" at u = 90°.

10.4 Ground stations and visibility

The stations come from **ground_stations.csv** (calculations._read_ground_stations). Columns:

Column	Unit	Empty	Use
`name, latitude_deg, longitude_deg, altitude_m`	deg, deg, m	required	Station position: geometry (WGS-84), ITU-R climate maps, Sun position, geomagnetic latitude.
`antenna_diameter_m`	m	field Antenna Diameter GS, else 3 m	D: ITU-R scintillation, G/T scaling, 3 dB beamwidth, Sun noise antenna model.
`gt_dbk`	dB/K	mission G/T	The station's own measured G/T (clear sky); replaces the mission G/T GS and its diameter scaling.
`gt_elevation_deg`	deg	mission G/T Reference Elevation (5° when that is empty too)	Elevation at which gt_dbk was measured.
`tsys_k`	K	mission System Noise Temperature	System noise temperature belonging to gt_dbk.

Column	Unit	Empty	Use
`antennas`		1	Antennas of the site, backups included. The statistics use 1 per site unless Use backup antennas is ticked.

Minimum elevation (Observation Settings, key `min_elevation_deg`, empty: 5°, allowed −5° to 90°) and the optional **horizon mask** (`horizon_masks/<station>.txt` or .csv: azimuth from North clockwise and minimum elevation, one point per line, interpolated linearly with wrap-around at North) give the visibility of each sample:

$$\mathrm{el}_{\mathrm{req}}(t) = \max\left[\mathrm{el}_{\min},\ m(\mathrm{az}(t))\right]$$

$\mathrm{el}_{\min}$: Min Elevation; m: horizon mask (0 without one).

$$\mathrm{contact}(t) \Leftrightarrow \mathrm{el}(t) \geq \mathrm{el}_{\mathrm{req}}(t)$$

A **contact** is a run of consecutive samples in contact. In the pass list it starts at its first sample in contact and ends at the first sample out of contact (or at the last sample of the day); **A/D** (ascending/descending) compares the sub-satellite latitude at the start and the end. In the statistics a contact of k samples lasts k × 30 s.

10.5 Geometry along the passes

At every sample Skyfield gives the topocentric position of the satellite seen from the station (calculations.prepare_topocentric_data):

Output	Formula	Notes
Elevation, azimuth, slant range d	Topocentric altitude, azimuth and distance (SGP4 + WGS-84)	No refraction.
Off Boresight Angle η (°)	angle between $-r_{sat}$ (nadir) and $r_{GS} - r_{sat}$	The satellite antenna is assumed to point at nadir.
Doppler Shift (kHz)	$f_D = -\frac{\dot{r}}{c} f, \quad \dot{r} = \mathbf{v}_{\mathrm{rel}} \cdot \hat{\mathbf{r}}$	Downlink frequency; positive when the satellite approaches.
Doppler Rate (Hz/s)	$\dot{f}_D = -\frac{f \ddot{r}}{c}$ $\ddot{r} = \frac{\lvert\mathbf{v}_{\mathrm{rel}}\rvert^2 - \dot{r}^2}{r} + \hat{\mathbf{r}} \cdot (\mathbf{a}_{\mathrm{sat}} - \mathbf{a}_{\mathrm{GS}})$	$a_{sat} = -\mu\, r_{sat} / \lvert r_{sat}\rvert^3$, $a_{GS} = -\omega_E^2$ (x, y, 0): exact at each sample, also at the closest approach.
Sun Separation (°)	angle between the Sun and the satellite seen from the station	$\cos\theta = \sin e_1 \sin e_2 + \cos e_1 \cos e_2 \cos(a_1 - a_2)$

The **static** link budget has no orbit: it takes a circular orbit at the entered altitude h and a spherical Earth (calculations.slant_range_from_elevation, off_nadir_angle_deg):

$$d = \sqrt{(R+h)^2 - (R+h_{\mathrm{GS}})^2 \cos^2 \mathrm{el}} - (R+h_{\mathrm{GS}}) \sin \mathrm{el}$$

Slant range.

$$\eta = \arcsin\left[\frac{(R+h_{\mathrm{GS}})\cos \mathrm{el}}{R+h}\right]$$

Angle from the satellite nadir to the station.

10.6 Atmospheric attenuation

The slant-path attenuation of ITU-R P.618 is computed by the itur library (itur.atmospheric_attenuation_slant_path) at the station latitude, longitude and altitude, for the carrier frequency f, the elevation el, the percentage of time p and the antenna diameter D (scintillation aperture averaging, antenna efficiency 0.5):

$$p = 100 - \text{availability } [\%]$$

Static link budget: limited to 0.001–50 %.

$$A = A_g + \sqrt{(A_r + A_c)^2 + A_s^2}$$

ITU-R P.618-13 total slant-path attenuation.

- A_g: gas, ITU-R P.676, with the water vapour density and content of the ITU-R maps at max(1, p) %.
- A_c: cloud, ITU-R P.840 at max(1, p) %.
- A_r: rain exceeded p % of the time, ITU-R P.618 with $R_{0.01}$ of P.837 and the rain height of P.839.
- A_s: tropospheric scintillation, ITU-R P.618, only when Tropospheric scintillation is on.
- **Static link budget** (atmospheric_attenuation_contributions): **Clear Sky** column = A_g only; **Rain Faded** column = A with gas, cloud, rain and, when switched on, scintillation, at the elevation of the tab.
- **Pass analysis** (atmospheric_attenuation_vs_elevation): A (rain faded, at p) is computed once on the elevation grid 0.5, 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 15, 20, 25, 30, 40, 50, 60, 75, 90° and interpolated linearly at the elevation of each sample (below 0.5° the 0.5° value). Every sample therefore carries the attenuation exceeded p % of the time: the pass results are rain-faded, not clear-sky.
- If the ITU-R model fails (e.g. outside its maps), the value at 5° is used at every elevation.
- The ITU-R tropospheric scintillation method is given for el ≥ 5° and about 4–20 GHz; outside it the results show a note (it is off by default).

10.7 Ground station G/T

Inputs (Downlink tab, shared with the Static LB · Downlink tab):

Parameter	Key / unit	Empty	Use
G/T GS	`gt_gs_dbk [dB/K]`	required	G/T of the reference antenna at the reference elevation, clear sky.
G/T Reference Diameter	`gt_ref_diameter_m [m]`	no scaling	D_{ref} of the antenna the G/T refers to.
G/T Reference Elevation	`gt_ref_elevation_deg [deg]`	5	Elevation of G/T GS (and of Tsys).
System Noise Temperature	`tsys_ref_k [K]`	constant G/T	$Tsys_{ref}$ at the reference elevation, clear sky. Enables the elevation/weather model, the radome loss and the Sun noise.
Radome Loss	`radome_loss_db [dB]`	0	Subtracted from the G/T (only with a System Noise Temperature).

The G/T used at a sample or in a column is obtained in four steps (link_core.station_gt, calculations.ground_station_gt_dbk):

$$(G/T)_{\text{ref}} = (G/T)_{\text{GS}} + 20 \log_{10} \frac{D}{D_{\text{ref}}}$$

Step 1, without the station's own G/T (no scaling without D_{ref}); otherwise $(G/T)_{ref}$ = gt_dbk of the catalogue.

$$T_{\text{sky}}(A) = T_m(1 - 10^{-A/10}) + T_c\, 10^{-A/10}$$

Step 2: sky noise through an attenuation A [dB]; T_m = 275 K, T_c = 2.7 K.

$$T_{\text{sky, ref}} = T_{\text{sky}}(A_g(\text{el}_{\text{ref}}))$$

Clear sky (gas only, p = 50 %) at the reference elevation.

$T_{\mathrm{sky}} = T_{\mathrm{sky}}(A_g + A_c + A_r)$	At the sample or column; scintillation radiates no noise and is excluded.
$T_{\mathrm{sys}} = T_{\mathrm{sys,\,ref}} - T_{\mathrm{sky,\,ref}} + T_{\mathrm{sky}}$	Step 3: receiver and feed noise kept, sky noise replaced.
$G/T = (G/T)_{\mathrm{ref}} - 10\log_{10}\dfrac{T_{\mathrm{sys}}}{T_{\mathrm{sys,\,ref}}} - L_{\mathrm{radome}}$	
$G/T \leftarrow G/T - \Delta_{\mathrm{Sun}}$	Step 4, pass analysis only (section 10.8).

Why the G/T changes with the elevation. The G/T GS is a measured or specified value: the antenna gain G over the system noise temperature $Tsys_{ref}$ at one elevation, the reference elevation, in clear sky. The gain of the dish does not depend on where it points, but Tsys does: it is the noise of the receiver and feed, which is fixed, plus the noise of the sky the antenna looks at, T_{sky}. An atmosphere that attenuates the signal by A dB also radiates noise (the absorbed power is re-emitted as heat at about Tm = 275 K): the longer the path through the atmosphere, the higher A and T_{sky}. At low elevation the path is longest; rain and clouds add attenuation and noise too. The program therefore keeps the fixed part of Tsys, $Tsys_{ref} - T_{sky,ref}$, and adds the sky noise of the actual elevation and weather (step 3). The G/T changes by the ratio of the two system noise temperatures, as G is unchanged.

- At the reference elevation in clear sky $T_{sky} = T_{sky,ref}$, so $G/T = G/T_{ref}$ − radome loss: the radome loss is subtracted at every elevation, G/T GS being the value without radome.
- Below the reference elevation, or in rain, $T_{sky} > T_{sky,ref}$ and the G/T is lower; above it the G/T is slightly higher.
- The effect is larger for a low-noise receiver (small $Tsys_{ref}$) and at higher frequencies.
- **The reference elevation matters**: the same G/T GS specified at zenith instead of 5° or 10° describes a worse receiver (the low zenith sky noise leaves more of $Tsys_{ref}$ to the receiver), so the G/T at low elevation comes out lower. When it is not known, 5° (the usual specification) is assumed; the station catalogue can give each station its own (gt_elevation_deg).
- Without a System Noise Temperature, steps 2–4 are skipped: $G/T = G/T_{ref}$ at every elevation and weather (no radome loss, no Sun noise).
- The static tab refuses a $Tsys_{ref}$ not higher than $T_{sky,ref}$. Ground spill-over of the side lobes (ground noise picked up at low elevation) is not modelled.
- The static link budget computes it at its elevation, once for each column (clear: gas only; rain faded: gas + cloud + rain at p). The pass analysis computes it with the rain-faded attenuation on the elevation grid of section 10.6 and interpolates it at the elevation of each sample.

Example (invented station at 64° N, 20° E, 300 m, 3.7 m dish, 2.25 GHz): G/T GS = 13.5 dB/K and $Tsys_{ref}$ = 160 K at 10° clear sky, radome 0.3 dB, availability 99.9 % (p = 0.1 %). The clear-sky gas attenuation at 10° gives $T_{sky,ref}$ = 14.2 K, so the receiver and feed account for 145.8 K. Computed by the program while this manual was built:

El [°]	A clear [dB]	T_sky clear [K]	Tsys clear [K]	G/T clear [dB/K]	A rain [dB]	T_sky rain [K]	G/T rain [dB/K]	G/T clear, ref. at zenith
2	0.960	56.7	202.5	12.18	1.134	65.3	12.00	11.98
5	0.385	25.8	171.6	12.90	0.453	29.7	12.80	12.66
10	0.193	14.5	160.3	13.19	0.227	16.6	13.14	12.94
20	0.098	8.8	154.6	13.35	0.115	9.8	13.32	13.09
45	0.047	5.7	151.5	13.44	0.056	6.2	13.42	13.17

El [°]	A clear [dB]	T_sky clear [K]	Tsys clear [K]	G/T clear [dB/K]	A rain [dB]	T_sky rain [K]	G/T rain [dB/K]	G/T clear, ref. at zenith
90	0.034	4.8	150.6	13.46	0.040	5.2	13.45	13.20

For example at 5° in clear sky: $T_{sky} = 275\,(1 - 10^{-0.385/10}) + 2.7 \cdot 10^{-0.385/10} = 25.8$ K; Tsys = 160 − 14.2 + 25.8 = 171.6 K; G/T = 13.5 − 10·log(171.6 / 160) − 0.3 = 12.90 dB/K. The last column takes the same 13.5 dB/K and 160 K as if specified at zenith ($T_{sky,ref} = 4.7$ K): 12.66 dB/K at 5°, 0.23 dB less.

10.8 Sun noise

Active in the pass analysis when **Sun noise** is ticked (key `sun_noise`, empty: on) and a System Noise Temperature is given (calculations.sun_noise_temperature_k). **Solar Flux** (key `solar_flux_sfu`, sfu = 10^{-22} W/m²/Hz, empty: quiet Sun) is the flux density at the downlink frequency.

$$n = \mathrm{JD} - 2451545.0$$

Days from J2000.0 (Astronomical Almanac, about 0.01°).

$$L = 280.460^\circ + 0.9856474^\circ\, n, \quad g = 357.528^\circ + 0.9856003^\circ\, n$$

Mean longitude and mean anomaly of the Sun.

$$\lambda_\odot = L + 1.915^\circ \sin g + 0.020^\circ \sin 2g, \quad \varepsilon = 23.439^\circ - 0.0000004^\circ\, n$$

Ecliptic longitude and obliquity.

$$\alpha = \mathrm{atan2}(\cos\varepsilon \sin\lambda_\odot,\ \cos\lambda_\odot), \quad \delta = \arcsin(\sin\varepsilon \sin\lambda_\odot)$$

$$\mathrm{GMST} = 280.46061837^\circ + 360.98564736629^\circ\, n, \quad H = \mathrm{GMST} + \lambda_{\mathrm{GS}} - \alpha$$

Hour angle; elevation and azimuth from H, δ and the station latitude (no refraction).

$$T_b = \frac{S \cdot 10^{-22}\, \lambda^2}{2\, k\, \Omega_\odot}$$

Solar Flux S [sfu] given.

$$T_b = 120000\, f^{-0.75}\ \mathrm{K}$$

Quiet Sun, f in GHz.

$$\theta_{3\mathrm{dB}} = 70\,\lambda/D, \quad G_{\mathrm{peak}} = (G/T)_{\mathrm{ref}} + 10\log_{10} T_{\mathrm{sys,\,ref}}$$

θ: Sun separation from the antenna axis.

$$G(\theta) = G_{\mathrm{peak}} - 12\left(\frac{\theta}{\theta_{3\mathrm{dB}}}\right)^2$$

Main lobe, $\theta < \theta_{3\mathrm{dB}}$.

$$G(\theta) = \min\left[\max\left(G_{\mathrm{main}},\ G_{\mathrm{side}}\right),\ G_{\mathrm{peak}}\right], \quad G_{\mathrm{side}} = \max\left(32 - 25\log_{10}\theta,\ -10\right)$$

$\theta \geq \theta_{3\mathrm{dB}}$: ITU-R S.465 side lobes (θ taken ≥ 1°).

$$\Delta T = \frac{T_b\, \Omega_\odot\, G(\theta)}{4\pi}$$

G linear; 0 with the Sun below −0.2665° of elevation.

$$\Delta_{\mathrm{Sun}} = 10\log_{10}\left(1 + \frac{\Delta T}{T_{\mathrm{sys}}}\right)$$

Sun Noise (dB): G/T and Eb/No drop by this value.

At 2.25 GHz the quiet Sun has T_b = 65,320 K. A pass is shown in orange in the contact table when the Sun noise reaches 0.5 dB; without a System Noise Temperature the table marks instead the passes with the Sun less than 5° from the satellite.

10.9 Downlink parameters

Fields of the Downlink tab (also shown, with blue labels, in the Static LB · Downlink tab). R_c is the coded bit rate, OH the overhead, F the formatting factor of the static tab.

Parameter	Key / unit	Empty	Use
Frequency	`frequency_ghz [GHz]`	required	f: free space loss, ITU-R attenuation, Doppler, λ for the beamwidth and the Sun noise, S4 scaling.
EIRP SAT	`eirp_sat_dbw [dBW]`	required	EIRP at the peak of the satellite antenna pattern, used when the static tab's EIRP source is 'EIRP value': EIRP(η) = EIRP SAT − (G_{peak} − G(η)).
Coded Bit Rate	`bitrate_mbps [Mbps]`	required	R_c, channel bit rate, coding included.
Overhead (coded / info)	`overhead`	required	OH = R_c / R_{info}: R_{info} = R_c / OH.
Roll-off Factor	`rolloff`	no PFD check	α: occupied bandwidth B = R_c (1 + α) / η_s.
Spectral Efficiency	`spectral_efficiency_bpshz [b/s/Hz]`	no PFD check	η_s (QPSK: 2).
Demodulator Loss	`demod_loss_db [dB]`	required	Subtracted from C/No.
Link Availability	`link_availability_pct [%]`	required	p = 100 − availability for rain and cloud.
Other Attenuations	`other_attenuations_db [dB]`	0	Subtracted from the received power.
C/Io	`c_io_dbhz [dBHz]`	none	Pass analysis only: C/(No+Io) = $-10\cdot\log(10^{-C/No/10} + 10^{-C/Io/10})$.
G/T GS and the G/T model	`see 10.7`		
Antenna Diameter GS	`antenna_diameter_m [m]`	3	D of a station without diameter in the catalogue.
Solar Flux, Sun noise	`solar_flux_sfu, sun_noise`	quiet Sun, on	See 10.8.
Tropospheric scintillation	`(switch)`	off	Adds A_s to the attenuation (10.6).
Include the losses entered in the Static LB · Downlink tab	`(switch)`	on	Takes from the static tab: the Tx chain, the ionospheric, polarisation and multipath losses, the modulation degradation, the formatting factor and the PFD limits (10.12).

10.10 Uplink parameters

Parameter	Key / unit	Empty	Use
Uplink Frequency	`uplink_frequency_ghz [GHz]`	no uplink	f_{UL}: uplink free space loss and attenuation.
EIRP GS	`eirp_gs_dbw [dBW]`	no uplink	Ground station EIRP (the station tracks: no pointing loss).
G/T SAT	`gt_sat_dbk [dB/K]`	no uplink	Satellite G/T at the antenna pattern peak; the pattern roll-off G_{peak} − G(η) is subtracted.
Uplink Bit Rate	`uplink_bitrate_mbps [Mbps]`	downlink R_c	$R_{c,UL}$.

Parameter	Key / unit	Empty	Use
Uplink Overhead	`uplink_overhead`	downlink OH	OH_{UL}.
Uplink Roll-off, Spectral Efficiency	`uplink_rolloff, uplink_spectral_efficiency_bpshz`		Only shown: information bit rate and bandwidth of the uplink in the Uplink tab.
Demodulator Loss SAT	`demod_loss_sat_db [dB]`	0	Subtracted from the uplink C/No.
Uplink Other Attenuations	`uplink_other_attenuations_db [dB]`	0	Subtracted from the uplink received power.

The uplink is computed only when Uplink Frequency, EIRP GS and G/T SAT are all given.

10.11 Pass analysis: the columns of a sample

For every sample above the horizon (calculations.calculate_link_budget_parameters; C/(No+Io) and Eb/No are meaningful in contact). L_{link} = ionospheric + polarisation + multipath losses of the static tab, D_{mod} its modulation degradation, F its formatting factor (each 0, 0 and 1 when the static losses are not included). T_{chain} = 10·log P_{Tx} − circuit loss − VSWR loss of the static tab.

Column	Formula	Notes
Path Loss (dB)	$L_{FS} = 20\log_{10} d + 20\log_{10} f + 92.45$	d in km, f in GHz; = 20 log(4π d / λ).
Atmospheric Att (dB)	A at the sample elevation	10.6, rain faded at p.
S/C Antenna Gain (dBi)	G(η): antenna_pattern.txt interpolated linearly	0 dBi outside the table.
S/C Antenna Loss (dB)	$\max\,[0,\ G_{peak} - G(\eta)]$	G_{peak}: highest gain of the pattern.
EIRP (dBW)	$T_{chain} + G(\eta)$ $EIRP_{SAT} - [G_{peak} - G(\eta)]$	First: static tab EIRP source 'Tx chain'; second: 'EIRP value'.
G/T GS (dB/K)	10.7 at the sample elevation, minus the Sun noise	
Link Losses (dB)	L_{link}	No receive pointing loss: the station tracks.
Rx Power (dBW)	$P_{Rx} = EIRP - L_{FS} - A - L_{other} - L_{link}$	
C/(No+Io) (dBHz)	$C/N_0 = P_{Rx} + G/T + 228.6 - L_{demod} - D_{mod}$ $C/(N_0 + I_0) = -10\log_{10}\left(10^{-\frac{C/N_0}{10}} + 10^{-\frac{C/I_0}{10}}\right)$	The second line only when C/Io is given.
Eb/No (dB)	$E_b/N_0 = C/(N_0 + I_0) - 10\log_{10}\frac{R_c F}{OH}$	Referred to the transfer frame bit rate.
Doppler Shift, Doppler Rate	10.5	The contact table gives the largest |shift| and |rate| of each pass.
PFD (dBW/m²/4kHz)	$EIRP - 10\log_{10}(4\pi d^2) - 10\log_{10}\frac{B}{4\,\mathrm{kHz}}$	d in m. When the static tab has a PFD limit and α, η_s are set.
PFD Limit, PFD Margin	limit(el) (10.14); margin = limit − PFD	Not shown in the tables (in the exported CSV); a pass or sample with a negative margin is shown in red.
UL Path Loss, UL Atmospheric Att	as above at f_{UL}	
UL Rx Power (dBW)	$P_{Rx,UL} = EIRP_{GS} - L_{FS,UL} - A_{UL} - L_{other,UL}$ $- [G_{peak} - G(\eta)] - L_{link,UL}$	Same satellite antenna pattern.

Column	Formula	Notes
UL C/No (dBHz)	$P_{\mathrm{Rx,\,UL}} + (G/T)_{\mathrm{SAT}} + 228.6 - L_{\mathrm{demod,\,SAT}} - D_{\mathrm{mod,\,UL}}$	
UL Eb/No (dB)	$C/N_{0,\,\mathrm{UL}} - 10\log_{10}\dfrac{R_{c,\,\mathrm{UL}}\,F_{\mathrm{UL}}}{OH_{\mathrm{UL}}}$	

10.12 Static link budget inputs

Keys of the sections `fixed_elevation_downlink` and `fixed_elevation_uplink` of parameters.json (written by **Save…**). The shared fields (frequency, EIRP, G/T, bit rate, overhead, roll-off, demodulator loss, other attenuations, availability, diameter) are the ones of 10.9 and 10.10.

Field	Key / unit	Empty	Use
Ground Station	`station`		Position for the ITU-R maps, the geomagnetic latitude and the G/T of the catalogue (independent of the pass analysis station).
Elevation Angle	`elevation_deg [deg]`	5	el of the budget (worst case: the minimum elevation).
Satellite Altitude	`sat_altitude_km [km]`	required	h: slant range and off-nadir angle.
Tropospheric scintillation	`scintillation`	off	A_s in the Rain Faded column.
Data framing	`framing_mode`	manual	'ccsds' derives OH and F (10.14).
RS Interleaving Depth I	`ccsds_interleaving`	5	1–8.
Frame OCF / FECF / Packet Error Control	`ccsds_ocf / ccsds_fecf / ccsds_packet_error_control`	on / off / on	4 B, 2 B, 2 B.
Source Packet Length	`ccsds_packet_length [B]`	not deducted	L_{pkt}.
Data Field Header (CUC)	`ccsds_packet_sec_header [B]`	0	Packet secondary header.
Formatting Factor	`formatting_overhead`	1	F = transfer frame bit rate / information bit rate.
Required Eb/No	`required_ebno_db [dB]`	no margin	Threshold for the Data Recovery Margin.
Required Nominal Margin	`required_margin_db [dB]`	3	Requirement on the Data Recovery Margin.
EIRP source	`eirp_mode`	EIRP value	'tx_chain' or 'eirp'.
Transmitter Power	`tx_power_w [W]`		P_{Tx} (Tx chain).
Antenna Circuit Loss	`ant_circuit_loss_db [dB]`	0	Cables, filters (Tx chain).
VSWR	`vswr [:1]`	no loss	Mismatch loss (Tx chain).
Antenna Gain	`ant_gain_dbi [dBi]`		Tx antenna gain (Tx chain).
Antenna gain from the pattern (DL)	`ant_gain_from_pattern`	off	Antenna Gain = G(η) at this elevation; the Tx pointing loss is then 0.

Field	Key / unit	Empty	Use
Tx Antenna Axial Ratio	`ant_axial_ratio_db [dB]`		AR_{Tx} for the polarisation loss.
Tx Antenna Pointing Loss	`tx_pointing_loss_db [dB]`	0	Subtracted from Rx Power.
Ionospheric loss mode	`ionospheric_mode`	manual	's4' computes it from S4 (10.14).
Ionospheric Loss	`ionospheric_loss_db [dB]`	0	Manual value.
Ionospheric S4 (reference)	`iono_s4_ref`	0.3	S4 at the reference frequency.
S4 Reference Frequency	`iono_s4_ref_freq_mhz [MHz]`	1575.42	f_{ref} of S4.
Polarisation Mismatch Loss	`polarisation_loss_db [dB]`	from the axial ratios, else 0	Subtracted from Rx Power.
Multipath Losses	`multipath_loss_db [dB]`	0	Subtracted from Rx Power.
PFD Limit 0–5° El	`pfd_limit [dBW/m²/4kHz]`	no PFD check	Limit up to 5°.
PFD Limit 25–90° El	`pfd_limit_high [dBW/m²/4kHz]`	0–5° value	Limit from 25°.
G/T SAT from the pattern (UL)	`gt_from_pattern`	off	G/T SAT − (G_{peak} − G(η)); the Rx depointing is then not used.
Modulation Degradation	`modulation_degradation_db [dB]`	0	Added to the demodulator loss.
Rx Antenna Axial Ratio	`rx_axial_ratio_db [dB]`		AR_{Rx}.
Rx Antenna Depointing Angle	`rx_depointing_deg [deg]`	no loss	θ_p.
Rx Antenna 3dB Beamwidth	`rx_beamwidth_deg [deg]`	70 λ / D (DL)	θ_{3dB} (required on the uplink with a depointing angle).
Tolerances	`tolerances.<key>.favourable_db, adverse_db, distribution`	none	Keys tx_power, tx_gain, propagation, rx_gt, rx_degradation (10.14).

10.13 Static link budget: formulas of each result

calculations.calculate_fixed_elevation_link_budget computes every row twice, Clear Sky and Rain Faded (10.6); C/Io is not used in the static budget.

Row	Formula	Notes
Tx Power [dBW]	$10\log_{10} P_{\mathrm{Tx}}$	P_{Tx} in W; Tx chain.
VSWR Loss [dB]	$-10\log_{10}(1-\Gamma^2), \quad \Gamma = \dfrac{\mathrm{VSWR}-1}{\mathrm{VSWR}+1}$	VSWR 1.5 → 0.177 dB.
Effective Gain [dBi]	$G_{\mathrm{eff}} = G_{\mathrm{ant}} - L_{\mathrm{circuit}} - L_{\mathrm{VSWR}}$	
EIRP [dBW]	$P_{\mathrm{Tx}} + G_{\mathrm{eff}}$	Tx chain; otherwise the EIRP field.

Row	Formula	Notes
Slant Range [km], Free Space Loss [dB]	$L_{FS} = 20\log_{10} d + 20\log_{10} f + 92.45$	d from 10.5, in km; f in GHz.
Gas / Cloud / Rain / Scintillation, Total Atmospheric Attenuation	10.6	Clear: gas only.
Ionospheric Loss [dB]	manual, or S4 fade (10.14)	
Polarisation Mismatch Loss [dB]	entered, or the co-polar value (10.14)	
Total Propagation Loss [dB]	$L_{prop} = L_{FS} + A + L_{iono} + L_{pol} + L_{mp} + L_{other}$	
Rx Antenna Pointing Loss [dB]	$12\left(\frac{\theta_p}{\theta_{3dB}}\right)^2$	Parabolic main-lobe approximation.
Rx Power [dBW]	$P_{Rx} = \mathrm{EIRP} - L_{prop} - L_{point,\,Tx} - L_{point,\,Rx}$	
Rx G/T [dB/K]	10.7 for the condition (clear or rain sky noise)	Uplink: G/T SAT (− roll-off).
Received S/No [dBHz]	$S/N_0 = P_{Rx} + G/T + 228.6$	
C/No after Degradations [dBHz]	$C/N_0 = S/N_0 - L_{demod} - L_{mod}$	
Transfer Frame Bit Rate [dBHz]	$10\log_{10} R_{frame}, \quad R_{frame} = \frac{R_c\, F}{OH}$	
Received Eb/No [dB]	$E_b/N_0 = C/N_0 - 10\log_{10} R_{frame}$	
Data Recovery Margin [dB]	$M = E_b/N_0 - (E_b/N_0)_{req}$	Green when ≥ Required Nominal Margin.
Margins with tolerances	10.14	MEAN − 3σ and Worst Case RSS ≥ 0 dB.
Power Flux Density [dBW/m²]	$\mathrm{EIRP} - 10\log_{10}(4\pi d^2)$	d in m; no atmospheric credit (conservative).
Occupied Bandwidth [kHz]	$B = \frac{R_c\,(1+\alpha)}{\eta_s}$	
PFD in 4 kHz	$\mathrm{PFD} - 10\log_{10}\frac{B}{4\,\mathrm{kHz}}$	Uniform spectral density.
PFD Limit, PFD Margin	$M_{PFD} = \mathrm{limit(el)} - \mathrm{PFD}_{4kHz} \geq 0$	limit(el): 10.14.

10.14 Formulas of the derived losses and margins

CCSDS TM framing (calculations.ccsds_tm_overheads)

$L_{frame} = 223\,I, \quad L_{cb} = 255\,I$	Transfer frame and RS codeblock [bytes].
$L_{df} = L_{frame} - 6 - L_{SH} - 4_{OCF} - 2_{FECF}$	Data field: primary header 6 B, secondary header, OCF and FECF when selected.

$$L_{\text{user}} = L_{\text{df}} \frac{L_{\text{pkt}} - 6 - L_{\text{CUC}} - L_{\text{PEC}}}{L_{\text{pkt}}}$$

With a Source Packet length; otherwise $L_{user} = L_{df}$.

$$F = \frac{L_{\text{frame}}}{L_{\text{user}}}, \quad c = 2\,\frac{L_{\text{cb}} + 4}{L_{\text{frame}}}$$

Formatting and coding factors (ASM 4 B, convolutional 1/2).

$$OH = F\,c, \quad R_{\text{frame}} = \frac{R_c}{c}, \quad R_{\text{info}} = \frac{R_c}{OH}$$

Ionospheric scintillation (calculations.ionospheric_scintillation_fade_db)

$$S_4(f) = S_{4,\,\text{ref}} \left(\frac{f}{f_{\text{ref}}} \right)^{-1.5}$$

ITU-R P.531, valid for S4 < 0.4.

$$P_{\text{pp}} = 27.5\, S_4^{1.26}\ \text{dB}, \quad L_{\text{iono}} = P_{\text{pp}}/2$$

Peak-to-peak fluctuation; the loss is half of it.

$$\sin\phi_m = \sin\phi \sin\phi_p + \cos\phi \cos\phi_p \cos(\lambda - \lambda_p)$$

Geomagnetic latitude; zones: equatorial $|\varphi_m| < 20°$, mid-latitude < 58°, auroral oval < 72°, polar cap above.

Polarisation mismatch (calculations.polarisation_mismatch_loss_db)

$$r_1 = 10^{AR_{\text{Tx}}/20}, \quad r_2 = 10^{AR_{\text{Rx}}/20}$$

Voltage axial ratios.

$$L_{\text{pol}} = -20 \log_{10} \frac{2\,(r_1 + r_2)}{(1 + r_1)(1 + r_2)}$$

Co-polar value (used).

$$L_{\text{best, worst}} = -10 \log_{10} \left[\frac{1}{2} + \frac{4 r_1 r_2 \pm (r_1^2 - 1)(r_2^2 - 1)}{2\,(1 + r_1^2)(1 + r_2^2)} \right]$$

Range shown for information.

PFD limit versus elevation (calculations.pfd_limit_at_elevation)

$$\text{limit(el)} = L_{\text{low}}$$

el ≤ 5°.

$$\text{limit(el)} = L_{\text{low}} + (L_{\text{high}} - L_{\text{low}})\,\frac{\text{el} - 5^\circ}{20^\circ}$$

5° < el < 25°.

$$\text{limit(el)} = L_{\text{high}}$$

el ≥ 25°; without L_{high}, L_{low} at every elevation.

Statistical margins (calculations.tolerance_statistics, margin_statistics)

$$a = -\text{adv}, \quad b = +\text{fav}$$

Limits of each tolerance, in dB of margin.

$$\text{TRI}: \quad \mu = \frac{a + b}{3}, \quad \sigma^2 = \frac{a^2 + b^2 - ab}{18}$$

$$\text{UNI}: \quad \mu = \frac{a + b}{2}, \quad \sigma^2 = \frac{(b - a)^2}{12}$$

$$\text{GAU}: \quad \mu = \frac{a + b}{2}, \quad \sigma^2 = \left(\frac{b - a}{6} \right)^2$$

Limits at ±3σ.

$$M_{\text{mean}} = M_{\text{nom}} + \sum_k \mu_k, \quad \sigma = \sqrt{\sum_k \sigma_k^2}$$

Independent contributions: means and variances add.

$$M_{-3\sigma} = M_{\text{mean}} - 3\sigma, \quad M_{\text{RSS}} = M_{\text{nom}} - \sqrt{\sum_k \text{adv}_k^2}$$ MEAN − 3σ and Worst Case RSS.

$$M_{\text{fav}} = M_{\text{nom}} + \sum_k \text{fav}_k, \quad M_{\text{adv}} = M_{\text{nom}} - \sum_k \text{adv}_k$$ All Favourable, All Adverse.

Tolerances: Transmitter Power, Tx Antenna Gain, Atmospheric / Propagation Loss, Rx G/T, Receiver Degradation, entered as positive dB of effect on the margin; default distributions TRI, TRI, GAU, TRI, TRI. They are applied to the Rain Faded margin in the With Tolerances page.

10.15 Worked example: static downlink

An invented S-band downlink: the station of 10.7 (64° N, 20° E, 3.7 m), el = 5°, h = 650 km, f = 2.25 GHz, availability 99.9 %; Tx chain 4 W, circuit loss 0.8 dB, VSWR 1.3, gain from the satellite antenna pattern; coded rate 4 Mbps with CCSDS framing I = 4, OCF; S4 0.2 at GPS L1; axial ratios 2 / 1 dB; multipath 0.2 dB; Rx depointing 0.1°; demodulator 1.5 dB, modulation 0.3 dB; required Eb/No 2.5 dB. Computed by the program while this manual was built; entered in the Static LB · Downlink tab they give the same values.

Step	Calculation	Value
CCSDS framing	frame 892 B, data field 882 B, codeblock 1020 B	F 1.01134, coding 2.29596, OH 2.32200
Off-nadir angle η	asin((R + h_GS) cos 5° / (R + h))	64.70°
Antenna gain G(η)	pattern interpolated	-0.46 dBi
Tx power	10·log 4 W	6.02 dBW
VSWR loss	Γ = 0.3/2.3; −10·log(1 − Γ²)	0.075 dB
EIRP	6.02 − 0.8 − 0.075 − 0.46	4.69 dBW
Slant range d	formula of 10.5	2447.3 km
Free space loss	20·log 2447.3 + 20·log 2.25 + 92.45	167.27 dB
Atmosphere clear / rain	gas / gas + cloud + rain	0.385 / 0.453 dB
Ionospheric loss	S4 0.2 → 0.117; 27.5·S4^1.26 / 2	0.92 dB
Polarisation loss	co-polar, AR 2 / 1 dB	0.057 dB
Rx pointing loss	12 (0.1 / 2.52)², θ3dB = 70 λ / D	0.019 dB
Rx power clear	EIRP − total propagation loss − pointing	-164.16 dBW
G/T clear	as in 10.7 at 5°: Tsys 171.6 K	12.90 dB/K
S/No clear	Rx power + G/T + 228.6	77.33 dBHz
C/No clear	− 1.5 − 0.3	75.53 dBHz
Transfer frame bit rate	4 Mbps / 2.32200 × 1.01134	1742.2 kbps = 62.41 dBHz
Eb/No clear / rain	C/No − 10·log R_frame	13.12 / 12.96 dB
Data Recovery Margin clear / rain	Eb/No − 2.5	10.62 / 10.46 dB
MEAN − 3σ (rain)	Tx power ±0.5 dB TRI, propagation ±1 dB GAU	9.28 dB
Worst Case RSS (rain)	margin − √(0.5² + 1²)	9.34 dB
PFD in 4 kHz	B = 2.7 MHz	-162.37 dBW/m²/4kHz
PFD margin	limit -154 at 5°	8.37 dB

11 Statistics reference

The Statistics tab repeats the pass analysis for every day of the period, every station of the network and every satellite, then adds up the samples (pass_statistics). The per-sample values (elevation, contact, Eb/No) are the ones of chapter 10, at 30 s steps.

11.1 Inputs

Field	Key / unit	Empty	Use
Start, Days	`[date], 1–366`	today, 7	Period, from 00:00 UTC of the start date.
Ground stations	`statistics_stations [list]`	none (required)	Network: a sample is usable through any of them.
Whole constellation	`(switch)`	on	Custom-orbit mission: every satellite of 10.3; else the satellite (plane, sat #) selected.
Required Eb/No	`(field) [dB]`	Static LB · Downlink value	Eb/No_{req}.
Required Margin	`(field) [dB]`	Static LB · Downlink value, else 0	M_{req}: a sample is usable when $Eb/No - Eb/No_{req} \geq M_{req}$.
Instrument Data, HKTM, Other Data	`downlink_data_kbps.instruments, .hktm, .other [kbps]`	not generated	Data stored on board; their sum is the generation rate G.
Real-time TM	`downlink_data_kbps.realtime [kbps]`	0	Sent live during the passes: the downlink rate for the stored data is R_{info} − real-time.
On-board Memory	`memory_gb [GB]`	unlimited	M_{max} = GB × $8 \cdot 10^9$ bit per satellite.
Min Pass Duration	`min_pass_min [min]`	0	Shorter usable contacts are not used.
Antenna Setup	`antenna_setup_min [min]`	0	Time to configure an antenna for another satellite.
Latency at end of pass	`latency_at_pass_end`	off	Data count as on ground when the pass ends.
Product Length	`product_length_min [min]`	none	P: the ground builds products of P minutes of on-board data; only complete products count (11.3).
Product Latency From	`product_latency_reference`	data	data: each datum's sensing; start / end: of the product's sensing.
Latency requirement, Within	`latency_requirement.minutes, .percent`	none, 100 %	Checked on the downloaded data.
Use backup antennas	`(switch)`	off	Use all the antennas of each site (column antennas), not one.

11.2 Usable samples and passes

$$m(s,g,t) = E_b/N_0\,(s,g,t) - (E_b/N_0)_{\text{req}}$$

s: satellite, g: station, t: sample.

$$\text{usable}(s,g,t) \Leftrightarrow \text{contact}(s,g,t) \;\wedge\; m(s,g,t) \geq M_{\text{req}}$$

$$C = R_d\,\Delta t, \quad R_d = \frac{R_c}{OH} - R_{\mathrm{RT}}$$

Capacity of a usable sample; Δt = 30 s; R_{RT}: real-time TM.

Min Pass Duration (drop_short_contacts): each run of usable samples of one satellite over one station, followed across midnight, shorter than ⌈duration / 30 s⌉ samples is removed (set out of contact) before anything else is counted.

One satellite (combine_stations, aggregate): at each sample the network is in contact when any station is; the station used is the one with the best margin. A pass is a run of samples in contact (one over midnight counts for the day it starts); its usable minutes are its usable samples × 0.5 min; the gap after a pass lasts until the next pass starts.

Several satellites (simulate_constellation): every antenna serves one satellite at a time and every satellite uses one antenna. At each sample:

- an antenna that served a satellite at the previous sample keeps it while that satellite stays usable there (a real antenna follows one pass);
- the free antennas are assigned by a maximum matching (augmenting paths), the satellites with more data on board first, each preferring its best-margin station; an antenna can take a satellite other than its last one only ⌈setup / 30 s⌉ samples after it stopped serving (Antenna Setup);
- a satellite that is usable somewhere but gets no antenna has a **conflict** for that sample;
- a site has 1 antenna, or its antennas column with Use backup antennas.

11.3 On-board memory and latency

Each satellite starts with an empty memory. At every sample t (aggregate, simulate_constellation):

$$Q \leftarrow Q + G\,\Delta t$$

Q: data on board; G = instruments + HKTM + other.

$$Q > Q_{\max}: \quad \mathrm{lost} \leftarrow \mathrm{lost} + Q - Q_{\max}, \quad Q \leftarrow Q_{\max}$$

Memory full: the new data are lost.

$$x = \min(Q,\ C)$$

Bits downloaded in the sample, when the satellite is served (else 0).

$$Q \leftarrow Q - x$$

The data leave the memory first in, first out. With D the bits downloaded before this sample, the x bits downloaded now were generated between D / G and (D + x) / G seconds from the start, uniformly. They reach the ground at t_{arr} = end of the sample, or, with **Latency at end of pass**, at the end of the pass (the run of served samples) they belong to. Each sample adds a chunk:

$$\ell_{\min} = t_{\mathrm{arr}} - \frac{D+x}{G}, \quad \ell_{\max} = t_{\mathrm{arr}} - \frac{D}{G}$$

Latency range of the chunk of x bits.

$$s(L) = \frac{\sum_k x_k\,\mathrm{clip}\left(\frac{60\,L - \ell_{\min,k}}{\ell_{\max,k} - \ell_{\min,k}},\ 0,\ 1\right)}{\sum_k x_k}$$

Share of the data on ground within L minutes.

$$L_X = \min\ \{L : s(L) \geq X\}$$

Latency at X % (bisection, 60 steps).

$$\bar{\ell} = \frac{\sum_k x_k \, (\ell_{\min,k} + \ell_{\max,k})/2}{\sum_k x_k}, \qquad \ell_{\max} = \max_k \ell_{\max,k}$$

Mean and maximum latency.

Ground products (Product Length P, pass_statistics._chunks): the ground processes the data in products of P minutes of on-board data, fixed blocks from 00:00 UTC (product k holds the data generated between kP and (k+1)P). A product can be processed only when all its data are on ground, at T_k, the arrival of its last chunk; products not complete at the end of the period are not counted. The chunks are split at the block boundaries and their latency is counted from:

$\ell = T_k - t_{\text{gen}}, \quad t_{\text{gen}} \in [kP,\ (k+1)P]$	Each datum's sensing (data): every datum waits for the whole product.
$\ell = T_k - kP$	Product sensing start (start): the strictest.
$\ell = T_k - (k+1)P$	Product sensing end (end).

With start or end all the data of a product have the same latency, so the share of the data is the share of the products (products of equal size).

So '80 % within 67 min' means: of all the bits generated on board and downloaded in the period (all satellites together for a constellation), 80 % reached the ground at most 67 min after they were generated. It is a share of the data volume, not of the time or of the passes. Data still on board at the end of the period are not counted. When the memory overflows the latency of the rest is approximate.

11.4 Results

Result	Definition	Notes
Passes, per day	number of passes (service windows per satellite for a constellation) / days	
Contact time [min/day]	samples in contact × 0.5 min / days	Served samples for a constellation.
Max / mean gap [min]	between the end of a pass and the start of the next	One satellite.
Usable time [min/day]	usable samples × 0.5 min / days	
Downlink capacity [MB/day]	usable samples × R_d · 30 s / 8·10⁶ per day (mean, min, max)	
Generated [MB/day]	G · 86400 / 8·10⁶ (× satellites)	
Downloaded [MB/day]	Σ x / 8·10⁶ per day	≤ capacity, ≤ generated.
On board [MB]	memory at the end of each sample (sum of all satellites); max and final	
Memory use [%]	100 · largest memory of one satellite / M_{max}	
Autonomy [h]	M_{max} / G / 3600: time to fill the memory without contacts	
Lost [MB, %]	Σ lost / 8·10⁶; 100 · lost / (G · period)	
Latency mean / max / at X %, within L	11.3 above	Requirement met when share(L) ≥ X %.
Per satellite: served, conflicts [min/day]	served / conflict samples × 0.5 min / days	

Result	Definition	Notes
Per station: served [min/day], volume [MB/day]	samples served by its antennas; data downloaded through it	
Per station: busy [%]	100 · served samples / (samples × antennas)	
Per station: contention [min/day]	samples with more usable satellites than antennas × 0.5 min / days	

11.5 Worked example: memory and latency

One satellite generating G = 60 kbps, information rate R_d = 1.5 Mbps (45 Mbit per sample), two contacts of 10 min from 60 to 70 min and from 160 to 170 min, 4 hours. Computed with pass_statistics.aggregate while this manual was built:

- At 60 min the memory holds 60 kbps × 3630 s = 217.8 Mbit (27.2 MB, the first sample's generation included); 45 Mbit leave per sample, so the backlog is gone in 5 samples (2.5 min), then each sample sends only the 1.8 Mbit generated in it.
- The oldest bit, generated at t = 0, lands at the end of the first contact sample (60.5 min): latency 60.5 min. The data generated during the contact land within a sample.
- Result: mean latency 36.1 min, max 90.5 min, 50 % within 34.5 min, 80 % within 59.0 min; 90.6 % within 75 min.
- With Latency at end of pass every chunk lands at 70 min (first pass) or 170 min (second): mean 43.8 min, max 100.0 min.

12 Data files

12.1 parameters.json

A JSON file with the values of the main tabs and two optional sections with the inputs of the static tabs (fixed_elevation_downlink, fixed_elevation_uplink, chapter 10.11). Every save writes all the keys below in this order, null when not used; a missing or null key leaves the field empty (e.g. c_io_dbhz: no interferer). Selecting another mission empties every field its file does not give. Mission Parameters... (Orbit & Passes) shows and edits the file as this table. All the keys:

Group	Key	Name [unit]	Meaning
Mission	`project_name`	Project name	Shown in the window title and in the reports.
	`norad_id`	NORAD ID	Catalogue number for the TLE download (launched satellites).
Orbit	`orbit.altitude_km`	Mean altitude [km]	Mean semi-major axis minus 6378.137 km.
	`orbit.eccentricity`	Eccentricity	Mean eccentricity.
	`orbit.arg_perigee_deg`	Argument of perigee [deg]	90 for a frozen orbit.
	`orbit.inclination_deg`	Inclination [deg]	Empty: sun-synchronous.
	`orbit.ltan`	Planes (LTAN) [hh:mm]	Local time of the ascending node of each plane.
	`orbit.satellites_per_plane`	Satellites per plane	Evenly spaced in the plane.
	`orbit.phasing_deg`	Phasing between planes [deg]	Shift of the argument of latitude from one plane to the next.
	`orbit.extra_satellites`	Extra satellites [LTAN@deg]	Single satellites added to the regular ones, e.g. 19:30@120 (plane, argument of latitude at epoch).
Downlink	`frequency_ghz`	Frequency [GHz]	Downlink carrier frequency.
	`eirp_sat_dbw`	EIRP SAT [dBW]	Satellite EIRP (when not from the Tx chain).
	`bitrate_mbps`	Coded bit rate [Mbps]	Channel bit rate, coding included.
	`overhead`	Overhead (coded / info)	Coded bit rate / information bit rate.
	`rolloff`	Roll-off factor	Occupied bandwidth = coded rate × (1 + roll-off) / spectral efficiency.
	`spectral_efficiency_bpshz`	Spectral efficiency [b/s/Hz]	2 for QPSK.
	`demod_loss_db`	Demodulator loss [dB]	Receiver implementation loss.
	`link_availability_pct`	Link availability [%]	Rain and clouds for p = 100 − availability.
	`other_attenuations_db`	Other attenuations [dB]	Any further fixed loss.
	`c_io_dbhz`	C/Io [dBHz]	Carrier to interference density; empty: no interferer.
Ground station	`gt_gs_dbk`	G/T GS [dB/K]	G/T of the reference antenna, reference elevation, clear sky.

Group	Key	Name [unit]	Meaning
	gt_ref_diameter_m	G/T reference diameter [m]	Antenna the G/T GS refers to.
	gt_ref_elevation_deg	G/T reference elevation [deg]	Elevation of the G/T GS.
	tsys_ref_k	System noise temperature [K]	At the G/T reference elevation, clear sky.
	radome_loss_db	Radome loss [dB]	Subtracted from the G/T.
	sun_noise	Sun noise	Sun in the antenna beam along the passes; empty: on.
	solar_flux_sfu	Solar flux [sfu]	For the Sun noise; empty: quiet Sun.
	antenna_diameter_m	Antenna diameter (default) [m]	For a station without a diameter in the station catalogue.
	min_elevation_deg	Minimum elevation [deg]	Minimum elevation of a contact.
Uplink	uplink_frequency_ghz	Frequency [GHz]	Uplink carrier frequency.
	eirp_gs_dbw	EIRP GS [dBW]	Ground station EIRP.
	gt_sat_dbk	G/T SAT [dB/K]	Satellite G/T.
	uplink_bitrate_mbps	Bit rate [Mbps]	Uplink coded bit rate.
	uplink_overhead	Overhead	Coded / information bit rate.
	uplink_rolloff	Roll-off factor	
	uplink_spectral_efficiency_bpshz	Spectral efficiency [b/s/Hz]	
	demod_loss_sat_db	Demodulator loss SAT [dB]	
	uplink_other_attenuations_db	Other attenuations [dB]	
Statistics	downlink_data_kbps.instruments	Instrument data [kbps]	Payload data stored on board and sent on this downlink.
	downlink_data_kbps.hktm	HKTM (stored) [kbps]	Housekeeping telemetry stored on board.
	downlink_data_kbps.other	Other data [kbps]	E.g. events and system logs.
	downlink_data_kbps.realtime	Real-time TM [kbps]	Sent live during the passes; takes part of the downlink bit rate.
	memory_gb	On-board memory [GB]	Memory of each satellite for these data.
	min_pass_min	Minimum pass duration [min]	Shorter contacts are not used.
	antenna_setup_min	Antenna setup [min]	Time to configure an antenna for another satellite.
	latency_at_pass_end	Latency at end of pass	Data on ground when the dump is complete.
	product_length_min	Product length [min]	Ground products of this many minutes of on-board data (fixed blocks from 00:00 UTC); empty: none.

Group	Key	Name [unit]	Meaning
	product_latency_reference	Product latency from	data (each datum's sensing), start or end (of the product's sensing).
	latency_requirement.minutes	Latency requirement [min]	
	latency_requirement.percent	Latency requirement, share [%]	
	statistics_stations	Ground stations	Network of the statistics.

The static sections are best written with **Save…** in the static tabs rather than by hand.

12.2 Other files

File	Format
tle.txt	The two TLE lines (an optional name line before them is accepted).
ground_stations.csv	CSV with the header name, latitude_deg, longitude_deg, altitude_m (required) and the optional columns (cells may be left empty): antenna_diameter_m; gt_dbk, the station's own measured G/T [dB/K, clear sky], which replaces the mission G/T GS scaled with the diameter; gt_elevation_deg, the elevation of that G/T (empty: the mission's G/T Reference Elevation, usually 5°: G/T values are normally given at 5°, clear sky; if it was measured at zenith instead, the low-elevation G/T is about 0.8 dB optimistic at L band); tsys_k (empty: the mission's); antennas (default 1), backups included: used by the statistics only with Use backup antennas. Excel's semicolon-separated CSV is accepted; the old ground_stations.csv without header is still read. Edit Stations... (Orbit & Passes) shows and edits it as a table.
antenna_pattern.txt	One point per line: angle from nadir [deg], gain [dBi], separated by a comma, tabs, spaces or a semicolon. It should cover the lowest elevation used (66° from nadir at 3° and 628 km).

13 Command line

lbtool.py runs the pass analysis and the statistics without the window, from the sources (Python environment of the project), for the missions of satellites.json and with the same calculations as the window. Examples:

- `python lbtool.py missions`: lists the missions.
- `python lbtool.py passes --mission <name> --station Svalbard --date 2026-10-14 -o passes.csv`: contacts of the day with maximum elevation, minimum Eb/No and PFD margin; the samples above the horizon go to the CSV file.
- `python lbtool.py statistics --mission <name> --stations Svalbard,Kangerlussuaq --start 2026-10-14 --days 30 --instruments 100 --hktm 4 -o stats.csv`: the statistics of the Statistics tab over a station network, with the on-board data rates in kbps.
- Options: `--download` (download the TLE first), `--plane hh:mm` (orbital plane of a custom-orbit mission), `--min-elevation`, `--required-ebno`, `--required-margin`, `--constellation`, `--latency` / `--latency-pct`; `-h` lists them all.

14 Troubleshooting

Symptom	Cause and remedy
The program stops with a licence message	See 2.2.
Fields empty at startup	parameters.json or tle.txt are not next to the executable (or not valid): copy them there, or load them from the Orbit & Passes tab.
'Ground stations: built-in list'	ground_stations.csv was not found: the built-in stations are used.
'Pattern: isotropic 0 dBi' (orange)	The mission folder has no readable antenna pattern: put an antenna_pattern.txt there.
Input Error message	A field is empty or not a number: the static tabs open its page and select it.
Static results marked OUTDATED	An input changed after Calculate: press Calculate again (exports are disabled until then).
No passes listed	No pass above 5° on that date, or the TLE is old: update tle.txt.